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1 Introduction

This report reflects on the design of a network connecting two office buildings in Astana and Almaty. The scenario required WAN connectivity, LAN segmentation, dynamic routing, and NAT. The made decisions with the reasoning throughout the project are explained in the sections below.

2 Network Design

2.1 Network Topology

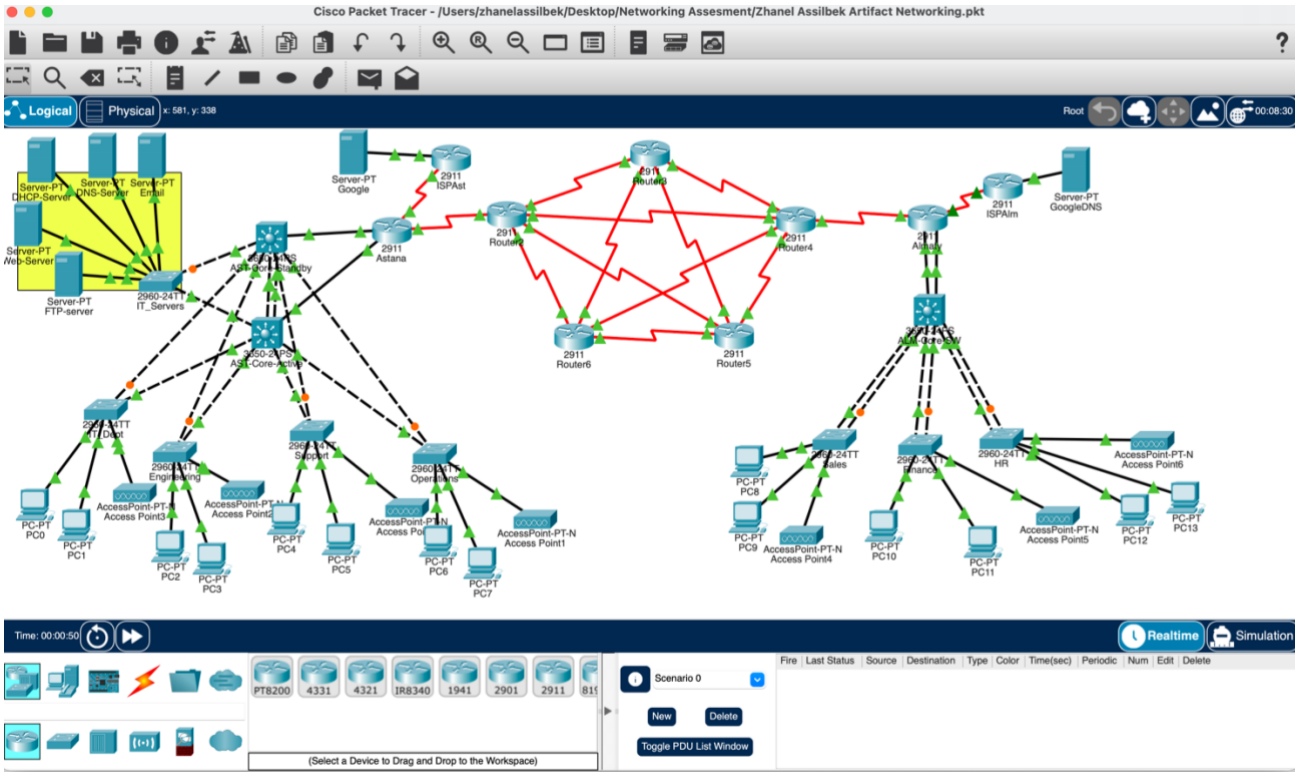


Figure 1 - Network Topology

A hierarchical star topology was used in both buildings because it keeps each department on its own access switch and suits VLAN segmentation(Varne et al., 2023). This makes the network easier to manage. For the WAN,partial mesh was chosen so that if one link fails, traffic can reroute another path(Sankpal et al., 2022). Full mesh topology was considered but rejected because of high cost and the need for extra links. Bus and ring were not selected because of single point of failure risk (Thakur & Verma, 2022).

2.2 Intermediate Network Design

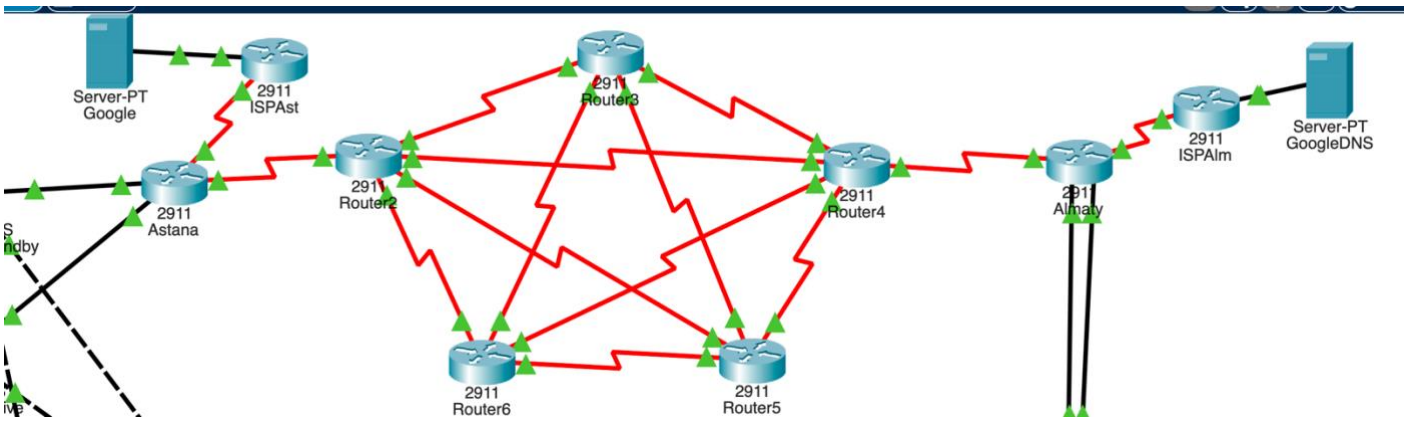


Figure 2 - Outline Intermediate Design

Six Cisco 2911 routers with ISP on each side create the WAN layer in a partial mesh. OSPF was chosen over RIP and static routing for specific reasons. Firstly, RIP is limited to 15 hops and converges slowly. Secondly, static routing needs manual updates after any change, which is time-consuming. OSPF is better option because it shares information about links across all routers and recalculates path automatically (Karamela & Karras, 2023; Shahid et al., 2024).

2.3 Local Network Designs

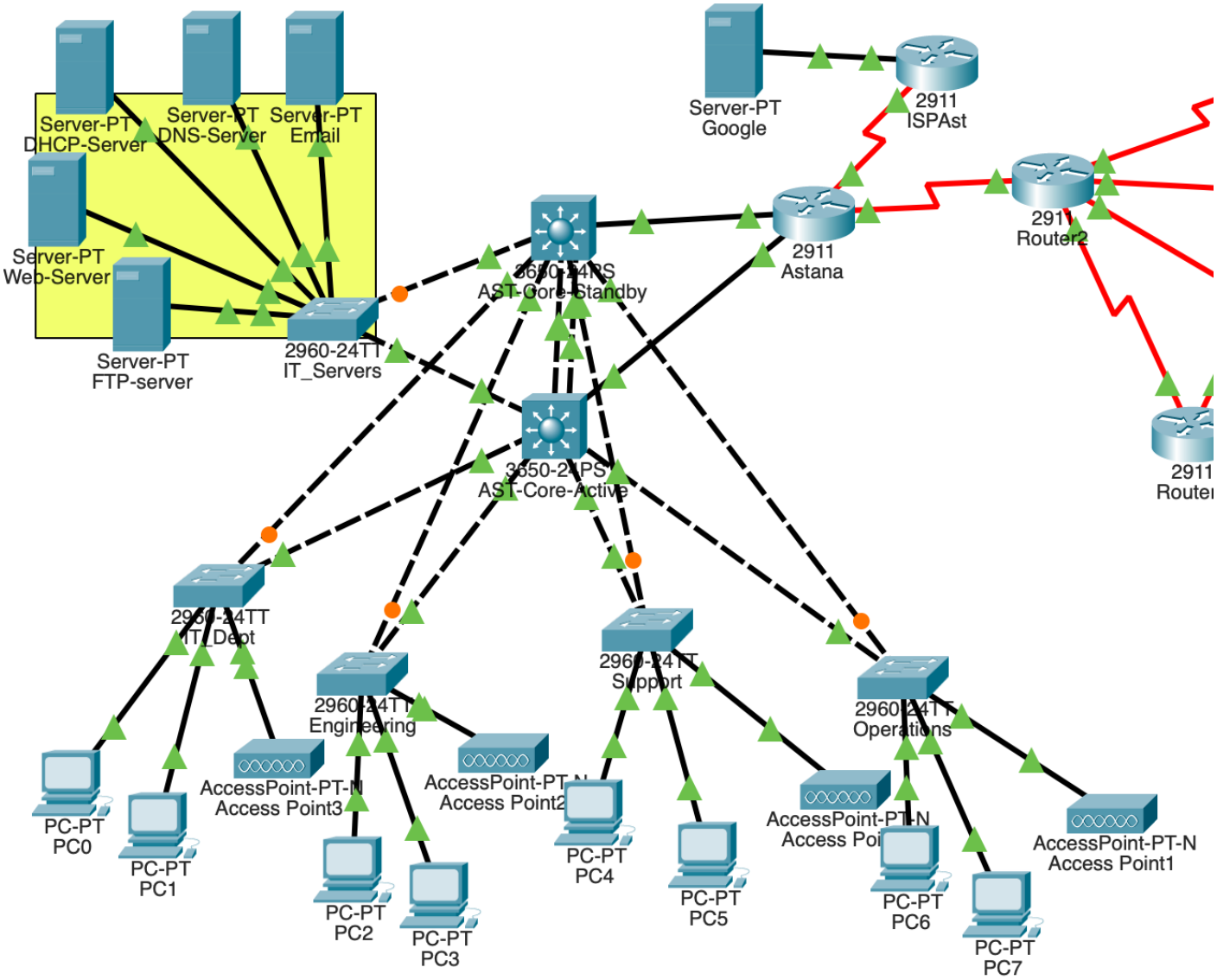


Figure 3 - Building A Local Network Design

Astana uses a hierarchical star topology with two L3 core switches: AST-Core-Active and AST-Core-Standby. They run HSRP to keep the gateway available if one switch fails. Four access switches serve IT, Engineering, Support, and Operations separately. Each of them connects to PCs and wireless access point. The server farm in VLAN 11 provides DHCP, DNS,FTP,Web and Email services to both buildings. Separating servers from users keeps the network organised and reduces broadcast.

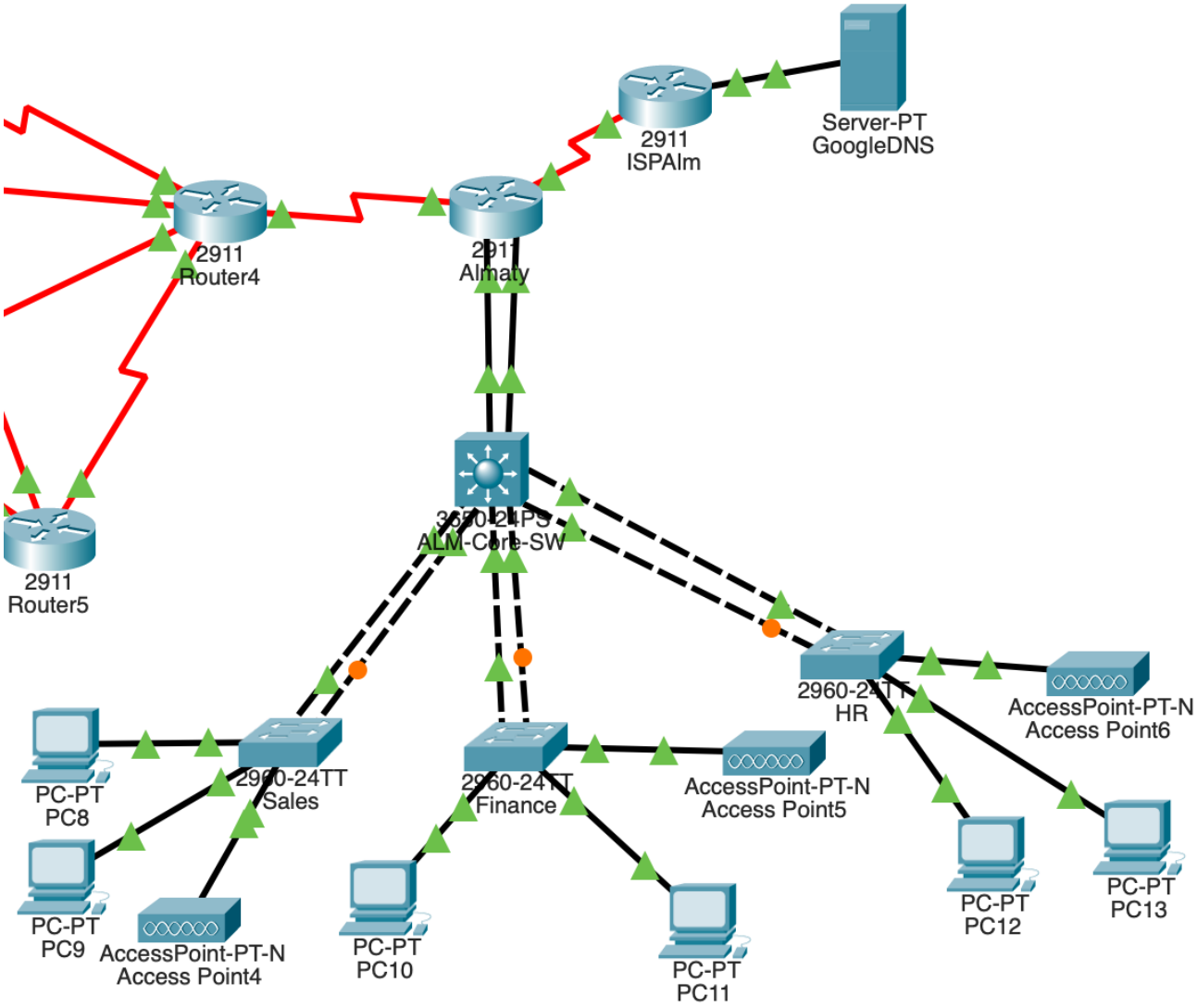


Figure 4 - Building B Local Network Design

Almaty uses a simpler structure with one core switch (ALM-Core-SW) that connects to three access switches for Sales, Finance and HR. Each of these L2 switches has PCs and wireless access point. Therefore, ALM-Core-SW has direct connection to the Almaty edge router, which handles both the WAN uplink and distributes DHCP addresses locally. A single core switch is enough for Almaty because it has fewer departments and lower traffic than Astana.

2.4 IP Addressing Scheme

Three address ranges were used: 172.16.10.0/23 for LANs, 10.10.22.0/24 for WAN point-to-point links, and 85.22.10.0/24 for public ISP connections. VLSM was applied to ensure that each subnet is sized to what the department requires without wasting addresses (Okoh et al., 2024). For example, a /26 gives 62 hosts for Astana's departments of 30 devices and considers 50% future growth. VLANs were assigned per department to separate traffic and simplify management. All the IP addressing tables are in Appendix B.

2.4.1 WAN IP Addresses

Every WAN link got a /30 subnet from 10.10.22.0/24. A /30 (2^2-2) gives exactly two usable addresses per point-to-point connection and avoids waste of addresses. Seven subnets cover all required connections including redundant paths such as Router2 to Router 6 and Router 5 to Router 4, which close the partial mesh loop and prevent connection failure. As a result, Astana and Almaty have constant connectivity even if primary link fails.

2.4.2 LAN IP Addresses (Building A)

Astana is divided into six VLANS using 172.16.10.0/23 range. IT_Dept (VLAN 10), Engineering (VLAN 20), Support (VLAN 30), and Operations (VLAN 40) each use a /26 (2^6-2) subnet with 62 usable hosts. This is enough for 30 current devices and 50% future growth. Each department switch also has PCS and a wireless access point to be able to connect more devices because L2 switch has only 24 ports. VLAN 99 with /27 (2^5-2 = 30) subnet handles remote access management by using SSH. VLAN 11 with /28 (2^4-2 = 14) subnet isolates important services from user devices. This makes network easier to control.

2.4.3 LAN IP Addresses (Building B)

Almaty uses three departmental VLANs from the same 172.16.10.0/23: Sales (VLAN 50), Finance (VLAN 60), and HR (VLAN 70). Each use a /27 (2^5-2) subnet with 30 usable addresses, covering 15 current devices with space for 50% growth. Finance and HR sit on separate VLANs because their data is more sensitive. VLAN 99 with /28 handles management access to the core switch via SSH.

2.5 Network Function

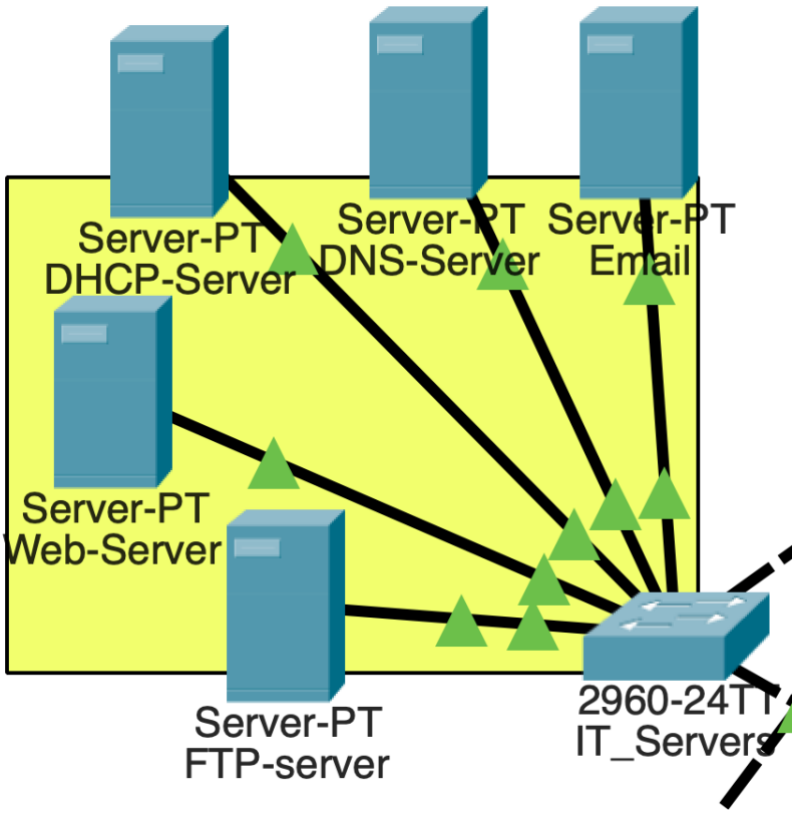


Figure 5 - Server Resources

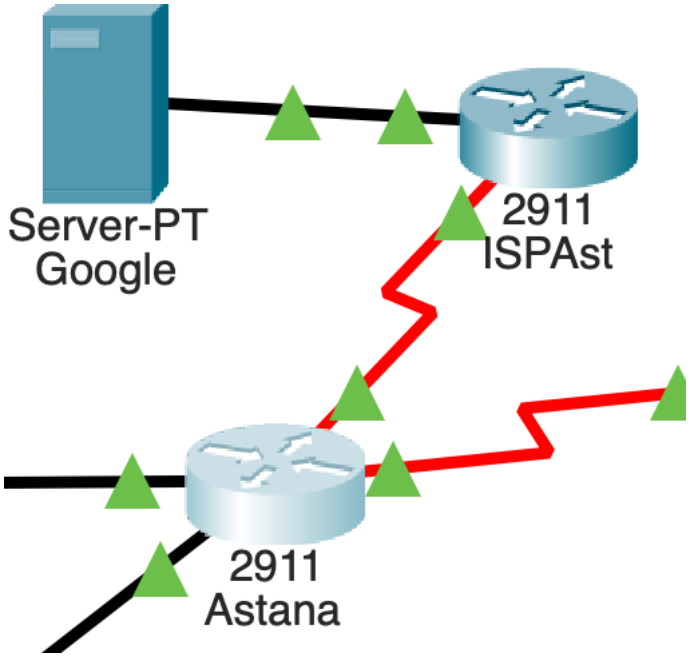


Figure 6 - Google simulation server in Astana

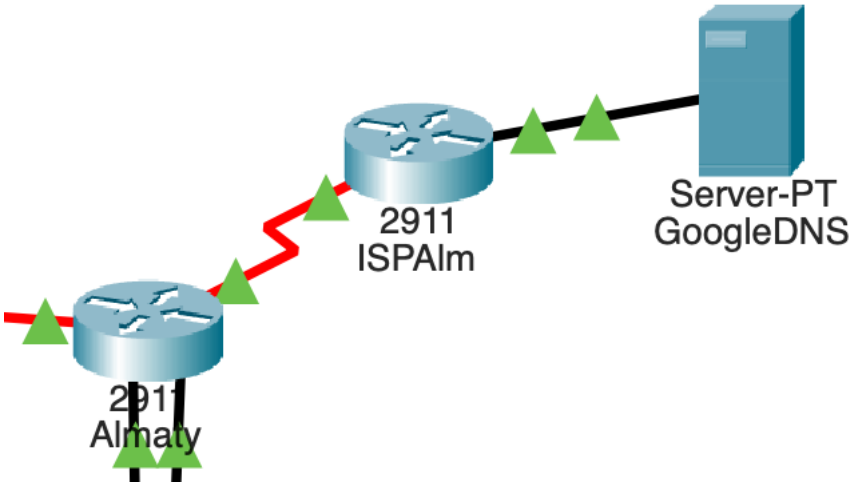


Figure 7 - Google simulation server in Almaty

Five PT servers run in VLAN 11(IT_Servers): DHCP, DNS, Web, FTP, and Email. PT Server was used because it supports all needed service types and static IP configuration. Centralizing services in Astana means that Almaty accesses them over the WAN rather than duplicating infrastructure. External servers simulate Google and internet access for testing. All the configuration information is in Appendix C.

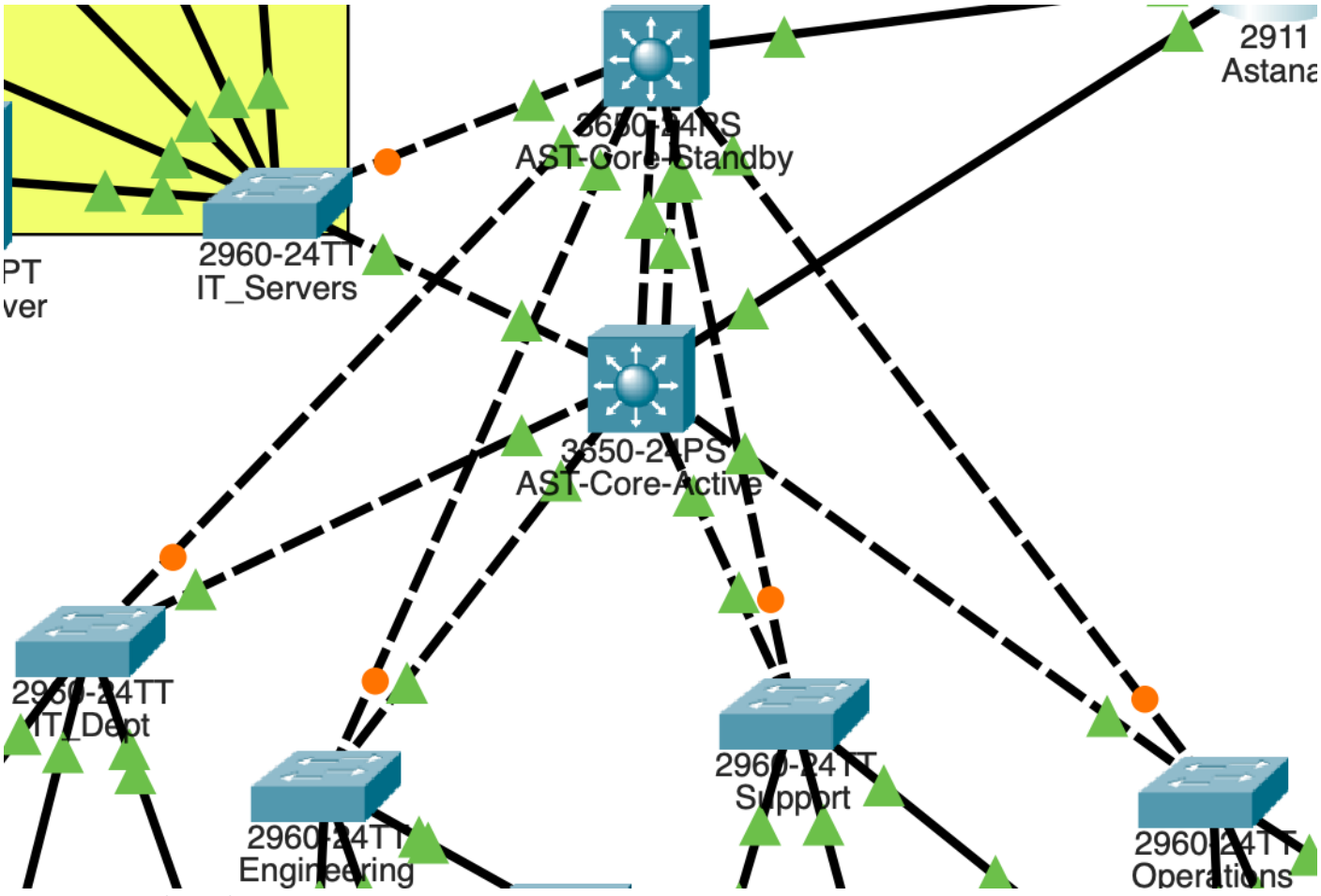


Figure 8 - Network Switches in Astana

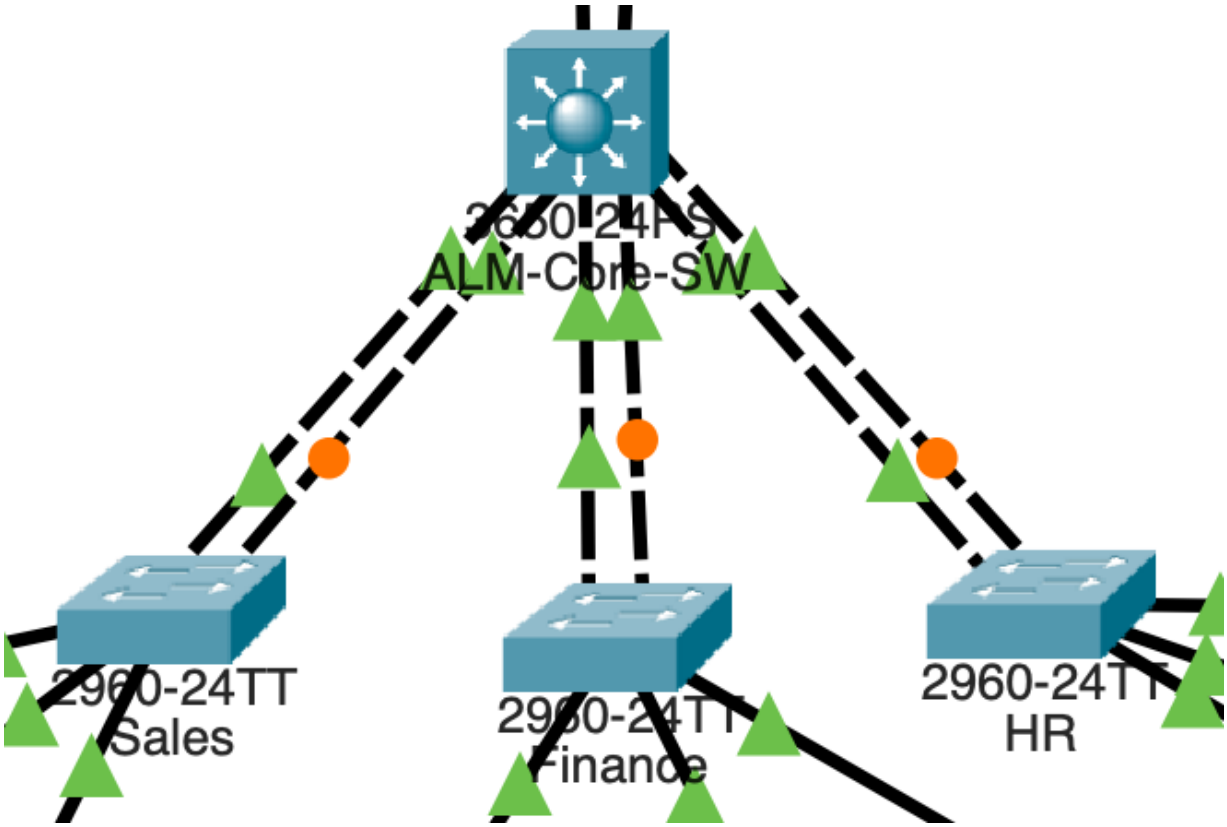


Figure 9 - Network Switches in Almaty

Two switch types were used. The Cisco 3650-24PS was chosen for core layer because it performs inter-VLAN routing without a separate router, which the 2960 cannot do. Moreover, 3650 switch has 28 GigabitEthernet ports, while 3560 switch has only 2 of them. This type of port is faster than FastEthernet. It also runs HSRP and Etherchannel at the same time without performance issues. In Astana, AST-Core-Active and AST-Core-Standby create a HSRP pair with Etherchannel grouping two physical links into one. This guarantees redundancy and high bandwidth. The Cisco 2960-24TT handles access layer connectivity in both buildings using 802.1Q trunk ports toward the core and access ports for each department. Rapid PVST+ works on all switches because it restores connectivity in a second after a change,while STP needs 30 to 50 seconds for this(Bhattarai, 2026).

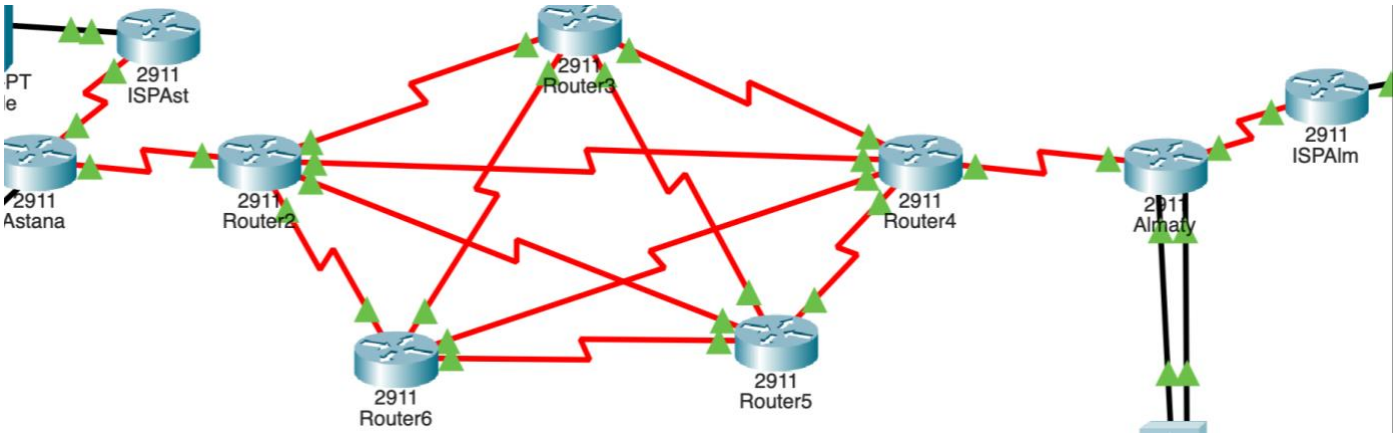


Figure 10 – Network Routers

Cisco 2911 routers were utilized for WAN and edge connections. The 2911 was selected because it supports OSPF and NAT overload without requiring extra hardware modules. Both the Astana and Almaty edge routers perform NAT overload, that allows local devices to share one public IP address to reach the internet. OSPF Area 0 runs across Router2 through Router6 and allows fast dynamic routing information exchange and best path selection (Karamela & Karras, 2023). ISPAst and ISPAlm use static routing since they only need a single default route.

3 Scalability, Security, Redundancy and Efficiency

3.1 Scalability

Each department has its own access switch. A single switch has only 24 ports, so it cannot accommodate multiple departments. Therefore, separating departments provides the ability to expand the network. Astana department runs on /26 subnets, which considers growth from 30 devices to 45 devices. Almaty department runs on /27 subnets, that fits raise from 15 to 23. As a result, less addresses are wasted. Wireless access points give a chance to connect even more devices (Varne et al., 2023). I configured fa0/1-20 as working ports for wired connection, which cover 20 devices. So, the remaining 10 plus 15 devices can be connected through access points in each Astana department. For Almaty, one switch is practical because it needs 23 physical ports considering future growth. So, other 3 devices can be connected through both wireless and wired connections. However, for a wired connection, the admin should do the “no shutdown” command on fa0/22-24 ports. In addition, I still connected access points in Almaty to provide variety of connections.

3.2 Redundancy & Efficiency

Reliance on a single core switch or one WAN link would have made the network weak. So, I used HSRP to ensure inter-VLAN and WAN communications. If AST-Core-Active stops to work, AST-Core-Standby will replace it until Active core switch recovers (Simanjuntak et al., 2023). EtherChannel between these two core switches provides both protection from connectivity failure and increased bandwidth. For stability, I implemented Rapid PVST+ on all switches for fast path recalculation, which prevent loops.

3.3 Security

```
IT_Dept#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)        (Count)
-------------------------------------------------------------------------
        Fa0/1        2           1                0         Restrict
        Fa0/2        2           1                0         Restrict
        Fa0/3        2           0                0         Restrict
        Fa0/4        2           0                0         Restrict
        Fa0/5        2           0                0         Restrict
        Fa0/6        2           0                0         Restrict
        Fa0/7        2           0                0         Restrict
        Fa0/8        2           0                0         Restrict
        Fa0/9        2           0                0         Restrict
       Fa0/10        2           0                0         Restrict
       Fa0/11        2           0                0         Restrict
       Fa0/12        2           0                0         Restrict
       Fa0/13        2           0                0         Restrict
       Fa0/14        2           0                0         Restrict
       Fa0/15        2           0                0         Restrict
       Fa0/16        2           0                0         Restrict
       Fa0/17        2           0                0         Restrict
       Fa0/18        2           0                0         Restrict
       Fa0/19        2           0                0         Restrict
       Fa0/20        2           0                0         Restrict
-------------------------------------------------------------------------
```

Figure 11 - Port Security example

```
interface FastEthernet0/22
 shutdown
!
interface FastEthernet0/23
 shutdown
!
interface FastEthernet0/24
 shutdown
!
```

Figure 12 - Shut down ports

```
AST-Core-Active#show access-lists
Standard IP access list ONLY_IT_ADMIN
    10 permit 172.16.10.0 0.0.0.63
    20 permit 172.16.11.0 0.0.0.31
    30 permit 172.16.10.160 0.0.0.15
    40 permit 172.16.11.160 0.0.0.15
    50 deny any

AST-Core-Active#
```

Figure 13 - ACL in Astana Core Switch

```
ALM-Core-SW#show access-lists
Standard IP access list ONLY_IT_ADMIN
    10 permit 172.16.10.0 0.0.0.63
    20 permit 172.16.11.0 0.0.0.31
    30 permit 172.16.10.160 0.0.0.15
    40 permit 172.16.11.160 0.0.0.15
    50 deny any

ALM-Core-SW#
```

Figure 14 - ACL in Almaty Core Switch

```
Astana#show access-lists
Standard IP access list 1
    10 permit 172.16.10.0 0.0.1.255
Standard IP access list ONLY_IT_ADMIN
    10 permit 172.16.10.0 0.0.0.63
    20 permit 172.16.11.0 0.0.0.31
    30 permit 172.16.10.160 0.0.0.15
    40 permit 172.16.11.160 0.0.0.15
    50 deny any

Astana#
```

Figure 15 - ACL in Astana Router

```
Almaty#show access-lists
Standard IP access list 1
    10 permit 172.16.11.0 0.0.0.255 (108 match(es))
Standard IP access list ONLY_IT_ADMIN
    10 permit 172.16.10.0 0.0.0.63
    20 permit 172.16.11.0 0.0.0.31
    30 permit 172.16.10.160 0.0.0.15
    40 permit 172.16.11.160 0.0.0.15
    50 deny any

Almaty#
```

Figure 16 - ACL in Almaty router

```
no profinet
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
!
!
!
no ip cef
ip routing
!
no ipv6 cef
!
!
!
username admin privilege 15 secret 5 $1$mERr$AFX/pZT1Lh7NP3Dp3P/qq/
username zhanel privilege 15 secret 5 $1$mERr$zDrUrgCFXW58wOFUOleAB.
!
!
!
!
!
!
!
!
!
!
ip ssh version 2
ip ssh time-out 60
ip domain-name zhcorp.kz
!
```

Figure 17 - Security measures: usernames and SSH

```
!
ip access-list standard ONLY_IT_ADMIN
 permit 172.16.10.0 0.0.0.63
 permit 172.16.11.0 0.0.0.31
 permit 172.16.10.160 0.0.0.15
 permit 172.16.11.160 0.0.0.15
 deny any
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 access-class ONLY_IT_ADMIN in
 login local
 transport input ssh
line vty 5 15
 access-class ONLY_IT_ADMIN in
 login local
 transport input ssh
!
!
```

Figure 18 - Security Configuration

The common security problems are unauthorised device plugging and uncontrolled traffic between departments. I applied Port security on all access switches by setting restrict violation mode and allowing only two MAC addresses for active user ports. Also, unused ports fa0/22-24 are shut down. I used VLAN segmentation to ensure that communication between VLAN will go through the core switch, because without it departments cannot collaborate (Bhattarai, 2026). To further improve security, ACLs between VLANs would limit which departments can communicate with specific servers or networks. Moreover, SSH and HTTPS protocols were integrated, because they are more secure than telnet and HTTP. The enable secret password(cisco) and usernames (admin-admin123; zhanel- zhanel123) also were set for high security.

4 Network Evaluation

4.1 Network Performance

The core switches and edge routers are connected through GigabitEthernet ports, which provides enough bandwidth for departments with 30 or more devices. Moreover, I configured EtherChannel between the two Astana core switches to further increase bandwidth. However, serial ports used in WAN links become a bottleneck because they are slower than Gigabit. Usage of /30 masks on WAN links reduces overhead and broadcast traffic (Okoh et al., 2024). Finally, OSPF improves performance by choosing the best available path (Shahid et al., 2024).

4.2 Design Challenges

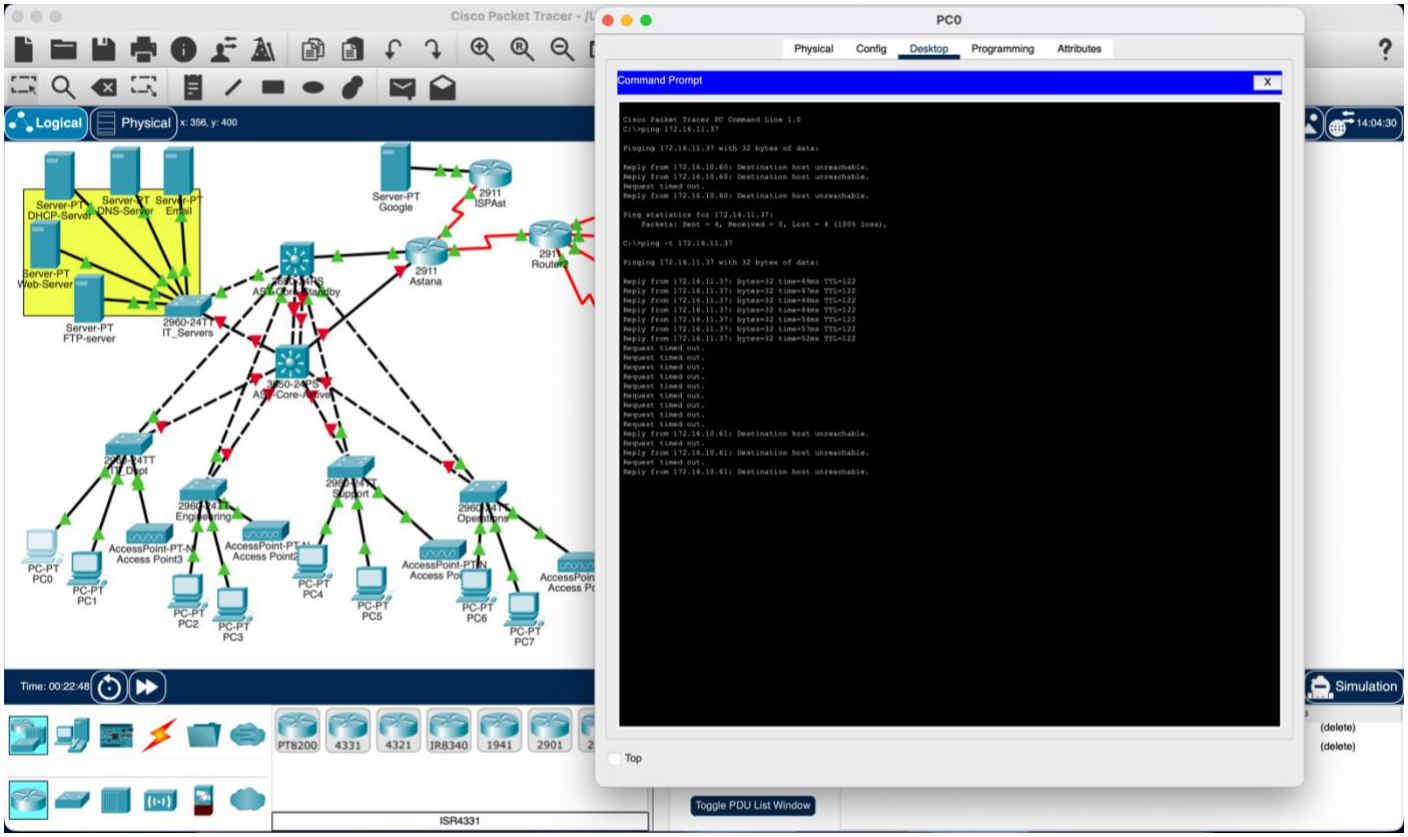


Figure 19 - Design Challenge

The hardest problem that I faced with was the Standby core switch being unable to reach Almaty while the Active switch had full connectivity. I found the issue by using these commands: show ip ospf neighbor and show ip route. OSPF was trying to use the virtual HSRP IP rather than the physical interface addresses, which caused asymmetric routing. I fixed it by redefining the network statements under the router ospf 1 to match the physical subnet interfaces. After that, successful ping and traceroute from the Standby switch demonstrated that the issue was resolved (Simanjuntak et al., 2023).

4.3 Testing Strategy

I started testing at the LAN level by checking communication between different VLANs within each building. Then I moved to test WAN connectivity between Astana and Almaty. Ping proofed that PCs from different cities can reach each other. Also traceroute verified that redundant paths worked when I manually shut down the primary link. Furthermore, I did some additional tests on servers and HSRP. Screenshots of the results are included in Appendix D.

5 Conclusion & Recommendations

In conclusion, I successfully connected two offices with a resilient network. I effectively used redundancy techniques such as HSRP and EtherChannel to protect Astana site from hardware and link failures. Challenge with asymmetric routing taught me that I should be careful with configurations because protocols can conflict with each other. I also learned to check CLI commands and analyze what mistakes I did . For future improvements, I recommend to add a second ISP link in Astana, implement HSRP in Almaty and create a guest VLAN to separate visitor traffic from constant users.

AI Use Category	Tool	How AI Tool was used in this assignment
Clarifying	Gemini	to clarify complex syntax for configuring HSRP
Verification	Gemini	to verify calculations for IP Addressing schemes
Editing	Gemini	to check grammar and improve language

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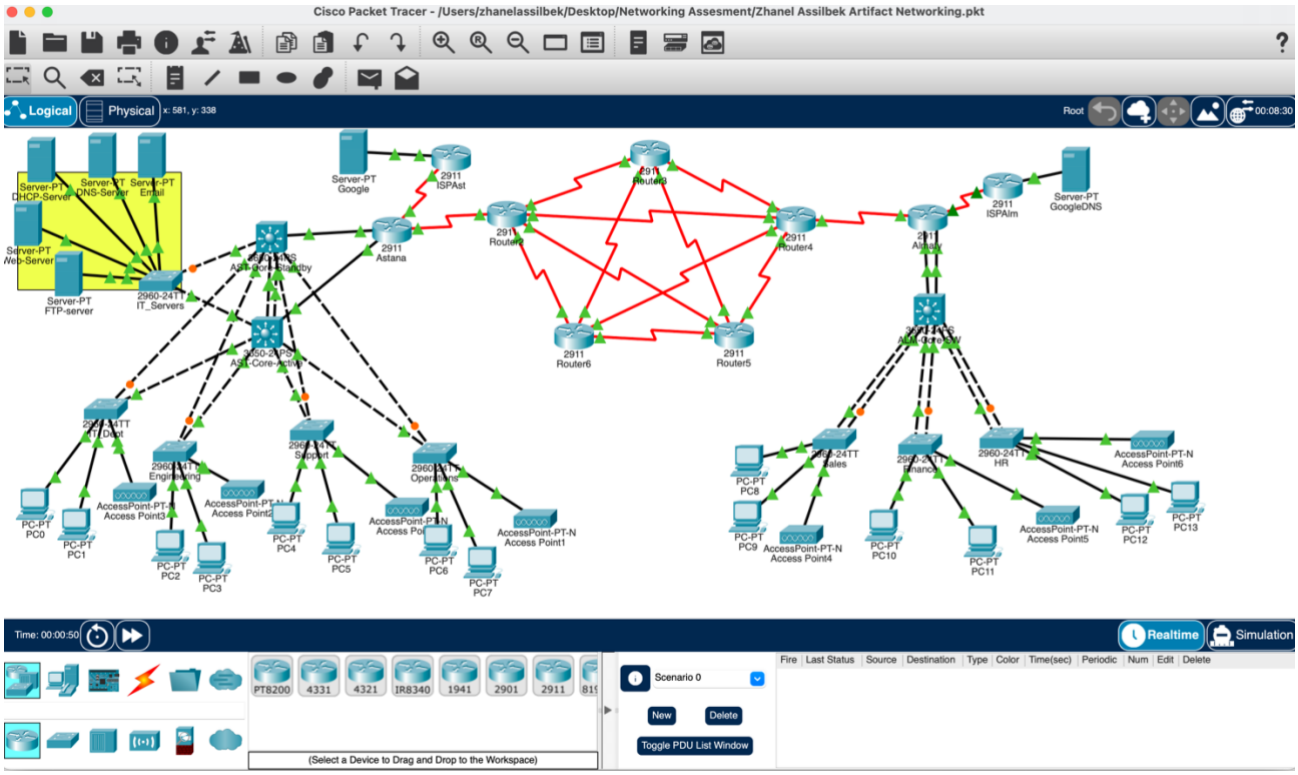
Simanjuntak, I. U. V., Haidi, J., Heryanto, & Silalahi, L. M. (2023). Simulation and performance analysis of network backup systems using Hot Standby Router Protocol (HSRP) method on real-time networks. *International Journal of Electronics and Telecommunications, 69*(4), 763–768. https://doi.org/10.24425/ijet.2023.147698

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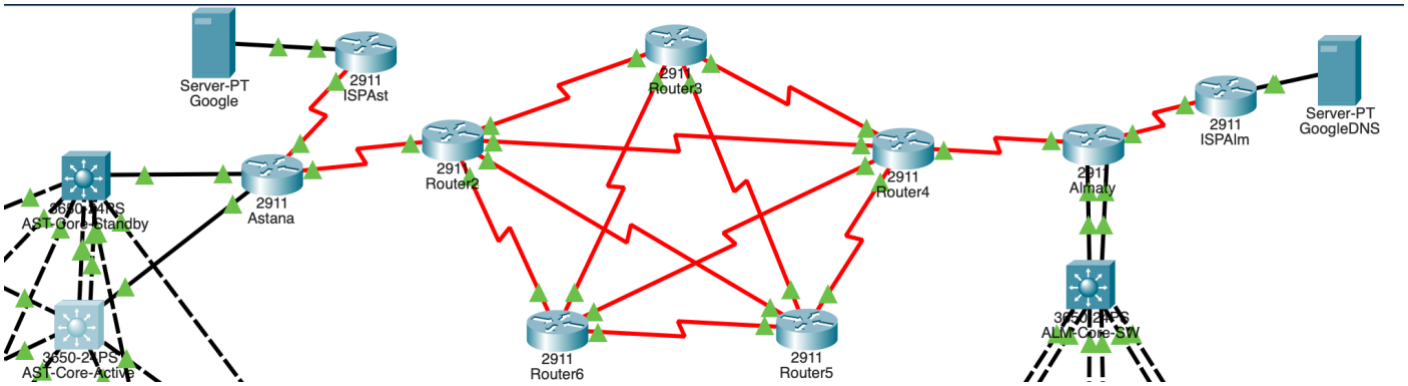
Varne, P. N., Priyanka, J., Shetty, V. S., Tejus, A. K., & Gowda, N. C. (2023). Campus network design and implementation using Cisco Packet Tracer. *International Journal of Computation Learning and Intelligence, 2*(4), 163–168. https://doi.org/10.5281/zenodo.10254264

Appendix A – Topologies

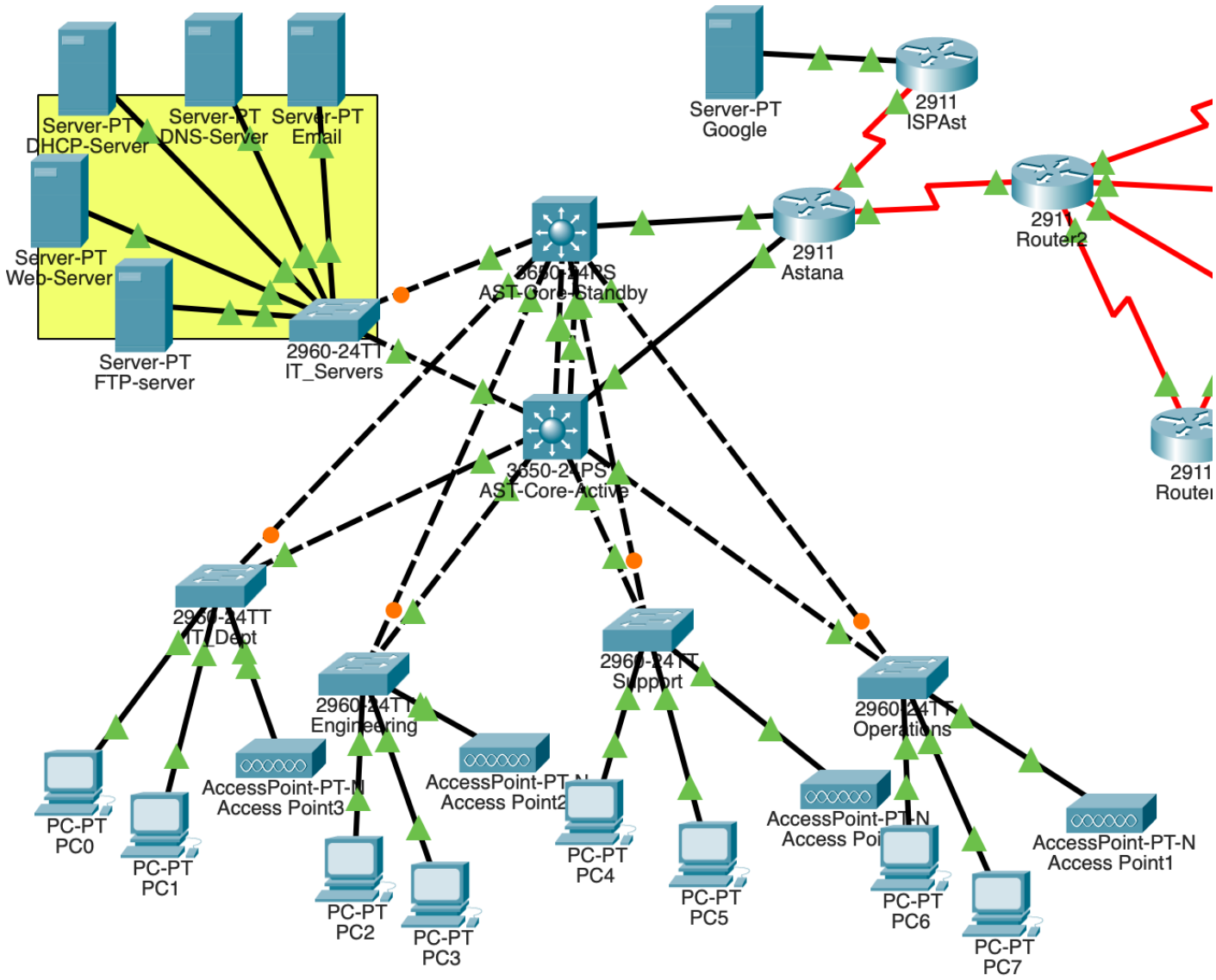
Topologies



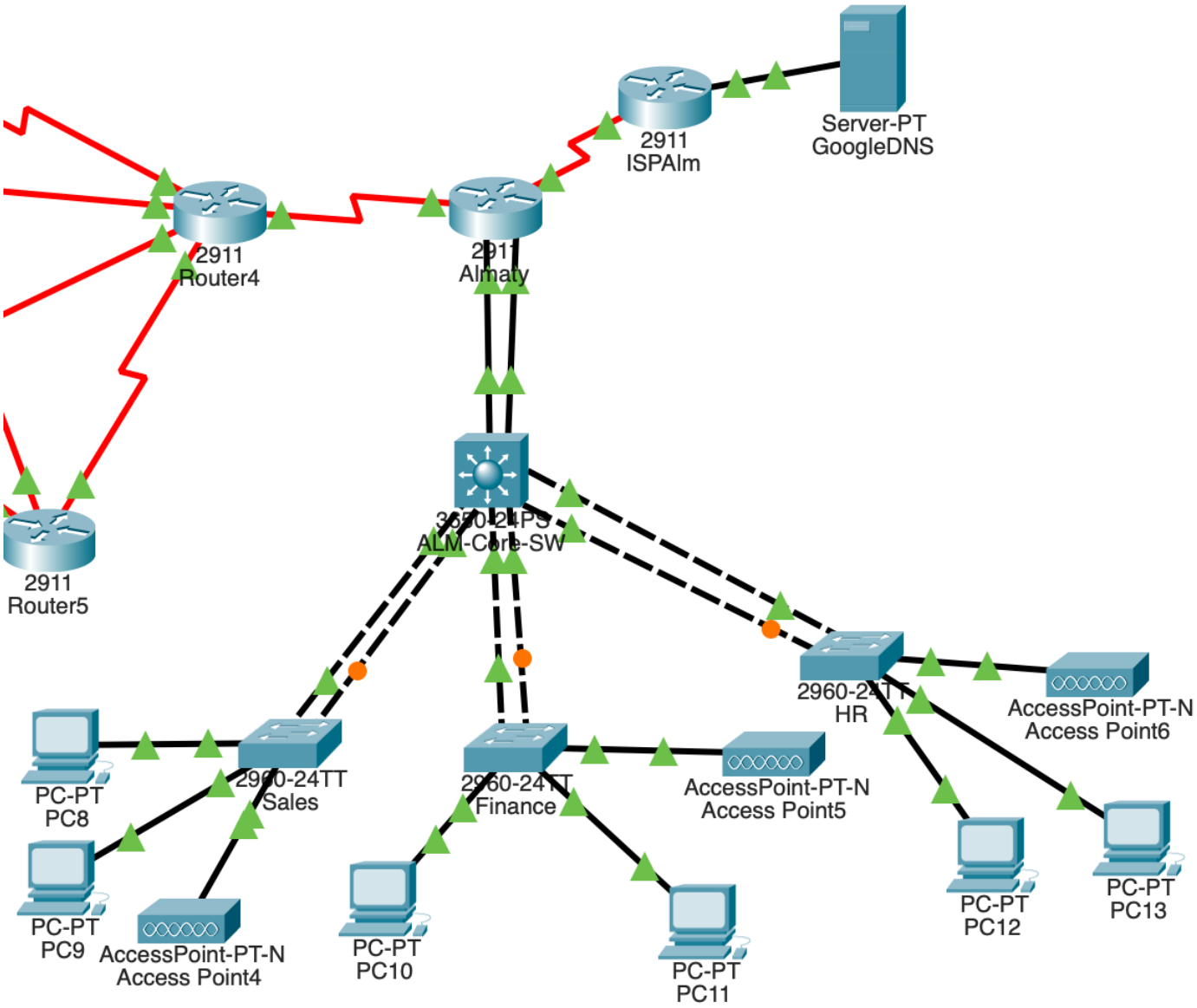
Intermediate Network



Local Network A



Local Network B



Appendix B - IP Addressing Scheme

Three separate address ranges cover the full network. The 172.16.10.0/23 handles all internal LAN segments in both buildings. The 10.10.22.0/24 range is used for all WAN point-to-point links between routers. The 85.22.10.0/24 provides public addresses for ISP connections. VLSM was applied throughout to size each subnet regarding to device requirements.

WAN IP Address Scheme

All WAN links use /30 subnets from the 10.10.22.0/24 range, giving two usable addresses per point-to-point connection. This keeps address waste minimal on links. Both the Astana and Almaty routers have /30 public subnets from the 85.22.10.0/24 range for their ISP uplinks. The 8.8.8.0 block simulates Google DNS to test internet reachability inside Cisco Packet Tracer.

Location	Network Address	Subnet Mask	Rationale
Astana-Router2	10.10.22.0	255.255.255.252	Connects the Astana edge router to the redundant WAN.
Router2- Router3	10.10.22.4	255.255.255.252	Redundant link between Router2 and Router3.
Router2-Router6	10.10.22.8	255.255.255.252	Redundant link between Router2 and Router6.
Router2-Router5	10.10.22.16	255.255.255.252	Redundant link between Router2 and Router5.
Router3-Router4	10.10.22.20	255.255.255.252	Bridging the central WAN segment
Router3-Router5	10.10.22.24	255.255.255.252	Redundant link between Router3 and Router5.
Router4-Router6	10.10.22.32	255.255.255.252	Redundant link between Router4 and Router6.
Router4-Router5	10.10.22.40	255.255.255.252	Redundant link between Router4 and Router5.
Router4-Almaty	10.10.22.44	255.255.255.252	Connects the redundant WAN to the Almaty edge router.
Astana-ISPAst	85.22.10.0	255.255.255.252	Gives the Astana branch internet connection
Almaty-ISPAlm	85.22.10.4	255.255.255.252	Gives the Almaty branch internet connection
External services	8.8.8.0	255.255.255.0	Simulate the Google DNS

Intermediate Level IP Address Scheme

The intermediate layer uses /30 subnets from the 172.16.11.0 range to connect the core switches to the edge routers. AST-Core-Active uses 172.16.11.252 on g1/0/9 as its primary uplink to the Astana router, while AST-Core-Standby uses 172.16.11.236 on g1/0/8 as a second physical path. Having two separate physical uplinks from the Astana router into the core switches follows the HSRP failover method. ALM-Core-SW follows the same pattern with primary and backup uplinks to the Almaty router.

Location	Network Address	Subnet Mask	Rationale

AST-Core-Active - Astana	172.16.11.252	255.255.255.252	Main path from building A to Astana via g1/0/9
AST-Core-Standby -Astana	172.16.11.236	255.255.255.252	Redundant physical link to Astana router via g1/0/8
ALM-Core-SW-Almaty(primary)	172.16.11.244	255.255.255.252	Main path from building B to Almaty via g1/0/7
ALM-Core-SW-Almaty (Backup)	172.16.11.240	255.255.255.252	Redundant physical link via g1/0/8

Building A (LAN) IP Address Scheme

The Astana LAN is divided into six VLANs. Four departmental VLANs use /26 subnets with 62 hosts, which is enough for current 30 and additional 15 devices. Management is VLAN 99 that uses a /27 subnet and controls remote access. VLAN 11 with a /28 subnet is isolates services such as DHCP and DNS from the rest of the network.

Device	Interface	IP Address[es]	Subnet	Gateway	Comment
IT_Dept	VLAN 10	172.16.10.1	255.255.255.192	172.16.10.1	Connects all IT staff devices
Engineering	VLAN 20	172.16.10.65	255.255.255.192	172.16.10.65	Network for engineering
Support	VLAN 30	172.16.10.129	255.255.255.192	172.16.10.129	Support staff and customers
Operations	VLAN 40	172.16.10.193	255.255.255.192	172.16.10.193	Handles general operations
Management	VLAN 99	172.16.11.1	255.255.255.224	172.16.11.1	Switch management and secure remote administration
IT_Servers	VLAN 11	172.16.11.129	255.255.255.240	172.16.11.129	Secure segment for servers

Building B (LAN) IP Address Scheme

The Almaty LAN uses three departmental VLANs from the same 172.16.10.0/23 range. Each use /27 subnets, which gives 30 usable addresses and covers the required 15 and future additional 8 devices. Finance and HR are on separate VLANs for higher level of security and scalability. VLAN 99 with a /28 subnet handles SSH management access to the Almaty core switch.

Device	Interface	IP Address[es]	Subnet	Gateway	Comment
Sales	VLAN 50	172.16.11.33	255.255.255.224	172.16.11.33	Connects all Sales department devices
Finance	VLAN 60	172.16.11.65	255.255.255.224	172.16.11.65	Connects all Finance department devices
HR	VLAN 70	172.16.11.97	255.255.255.224	172.16.11.97	Connects all HR department devices

Management	VLAN 99	172.16.11.161	255.255.255.240	172.16.11.161	Switch management and secure remote administration

Appendix C – Device Purpose and Configuration

Top Level (WAN)

At the WAN level, the Astana and Almaty routers are the main edge devices for each site. Both run OSPF and NAT. The Astana router uses serial interfaces for the WAN connection, while the Almaty router additionally executes DHCP for its local departments. ISPAst and ISPAlm use static routing to connect each site to the internet. Router2 through Router6 form the redundant WAN running OSPF Area 0. Google and GoogleDNS with static public IPs simulate internet services for testing. To protect the network, I disabled Telnet because it transmits data in plain text. Instead, I configured SSH version 2 on all routers, because it provides encryption of information. Also, I applied ACLs to the edge routers to ensure that only authorized traffic can pass between LAN and WAN links.

Device Type	Device Name	Configuration Information	Specific Purpose / Rationale
Router	Astana	OSPF, NAT, Serial, ACL, SSH	Acts as the Astana edge gateway; handles NAT establishes OSPF adjacencies over the WAN.
Router	Almaty	OSPF, DHCP, Serial, NAT, ACL, SSH	Acts as the Almaty edge gateway and DHCP server; manages dynamic routing and NAT
Router	ISPAst	Static Routing, NAT	Provides access to the Internet through NAT
Router	ISPAlm	Static Routing, NAT	Provides access to the Internet through NAT
Router	Router2, Router3, Router4, Router5, Router6	OSPF Area 0, Serial Interfaces, SSH	Core WAN Backbone routers that run OSPF and provide redundancy
Server	Google, GoogleDNS	Static IP configuration, Public WAN	Simulates Google website and access to the internet.

Intermediate Level

The intermediate layer is between the edge routers and the access switches. Cisco 3650-24PS Layer 3 switches were used in both buildings instead of the 3560 series. The main reason is that the 3650 can run inter-VLAN routing and other protocols without performance loss. In Astana, I used two switches that run HSRP: AST-Core-Active and AST-Core-Standby. So, if the Active switch fails, the Standby becomes active and takes control immediately. HSRP was not used in Almaty because fewer departments and traffic. In Almaty, a single ALM-Core-SW handles inter-VLAN routing and uses IP helper-address to forward DHCP requests up to the Almaty router. Both edge routers also appear at this level since they connect the intermediate and WAN layers. Furthermore, both the core switches and edge routers at this level are secured using SSH version 2 for encrypted management and ACLs to restrict unauthorized access.

Device Type	Device Name	Configuration Information	Specific Purpose / Rationale

Multilayer 3650 Switch	AST-Core-Active, AST-Core-Standby	Routed, HSRP, EtherChannel, 802.1Q, Spanning-tree mode rapid-pvst, ACL, SSH	Active: Primary core switch for building A Standby: Secondary core switch
Multilayer 3650 Switch	ALM-Core-SW	Routed, 802.1Q, Spanning-tree mode rapid-pvst, ACL, SSH	Core Switch for Building B
Router	Astana	OSPF, NAT, Serial, ACL, SSH	Connect building A to the WAN
Router	Almaty	OSPF, DHCP, Serial, NAT, ACL, SSH	Connect building B to WAN and acts as a DHCP server

Building A (LAN)

Four Cisco 2960-24TT switches handle access layer connectivity for IT_Dept, Engineering, Support, and Operations. Trunk ports configured for the core and access ports configured for end devices 802.1Q tagging. An IT_Servers switch runs on VLAN 11 with PortFast enabled to give servers a stable and fast connection. For security, I configured Port Security for fa0/1-20 ports in restrict mode on all L2 switches to prevent unauthorized access. Also, fa0/22-24 ports are shut down, because they are unused. PCs receive addresses via the DHCP server. Four wireless access points secured with WPA2 provide coverage across the Astana office, one per department switch via f0/21 port.

Device Type	Device Name	Configuration Information	Specific Purpose / Rationale
L2 Switch	IT_Dept, Engineering, Support, Operations	Trunk Port, Access Ports and VLANs, Spanning-tree mode rapid-pvst, Port Security	Access layer switches that manage physical connectivity and enforce logical segmentation through VLAN tagging.
L2 Switch	IT_Servers	VLAN 11, PortFast, Spanning-tree mode rapid-pvst, Port Security	Provides a fast and stable dedicated connection for all high-performance servers.
PC	PC0, PC1, …	DHCP IP configuration	End-users
Access Point	Access Point0, Access Point1, Access Point2, Access Point3	WPA2 Security, SSID	Provides secure wireless access and ensures signal coverage across the Astana office.
Server	DHCP-Server, DNS-Server, Web-Server,	Static IP Configuration	Provides essential network services such as IP addressing, name resolution, and web hosting.

	FTP-Server, Email-Server		

Building B (LAN)

Three access switches serve Sales, Finance, and HR in Almaty, each configured with the relevant VLANs 50, 60, and 70 using trunk and access ports. Like the Astana site, all L2 switches have Port Security configuration to prevent unauthorized access via restrict mode and shutdown of unused ports. PCs connect via DHCP, and three WPA2-secured access points provide wireless coverage across the Almaty office. The design is simpler than Astana since there is no server farm or dual-core setup, but the VLAN structure keeps departmental traffic isolated in the same way.

Device Type	Device Name	Configuration Information	Specific Purpose / Rationale
L2 Switch	Sales, Finance, HR	VLAN 50,60,70, Trunk, Access ports, Spanning-tree mode rapid-pvst, Port security	Provides localized access layer connectivity, ensuring departmental traffic remains isolated within the specified broadcast domains.
PC	PC8, PC9, …	DHCP IP configuration	End-users
Access Point	Access Point4, Access Point5, Access Point6	WPA2 Security, SSID	Provides secure wireless access and ensures signal coverage across the Almaty office.

Appendix D - Testing Documents

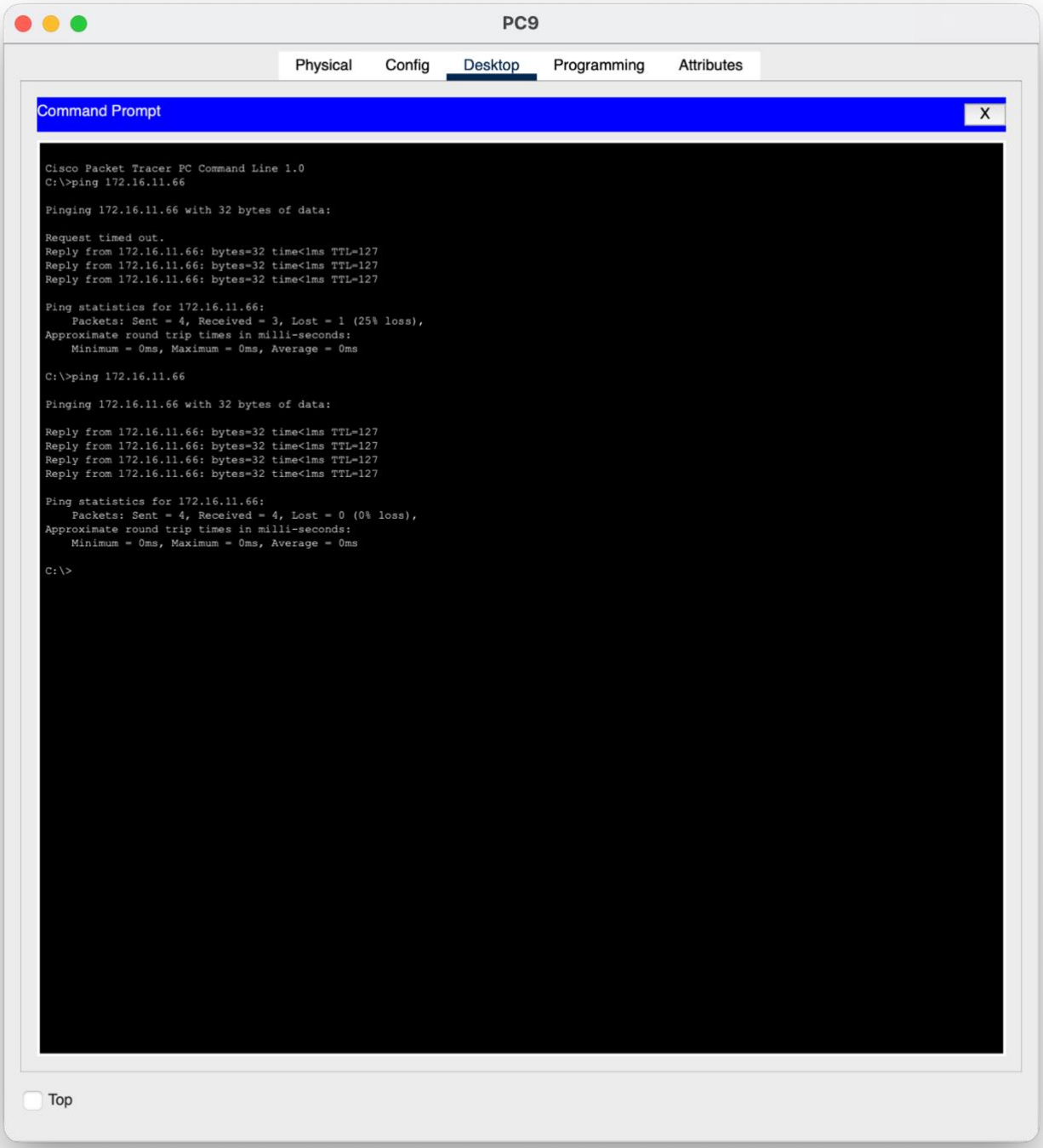


Figure D1 - Successful ping from Sales to Finance department in Almaty

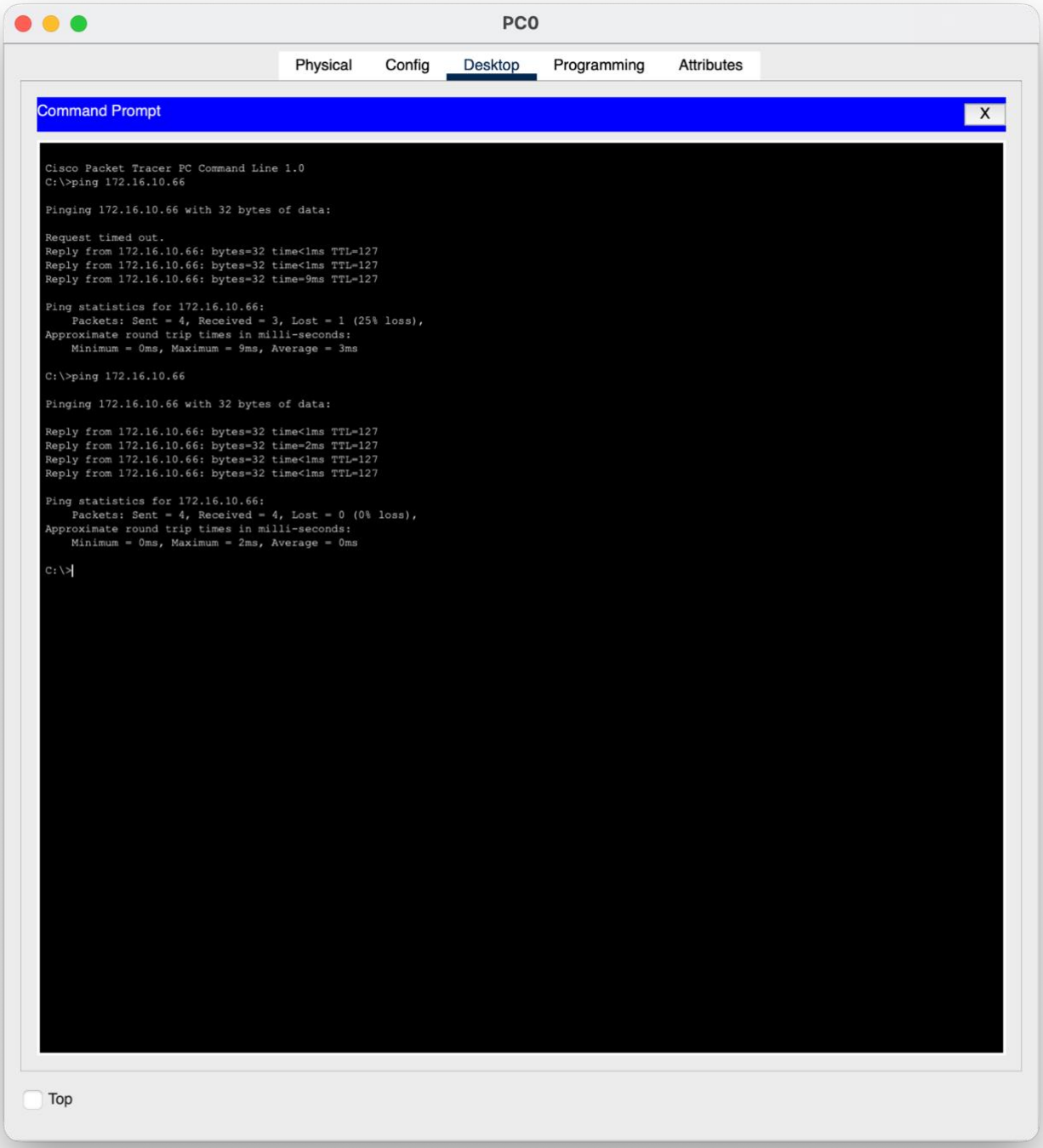


Figure D2 - Successful ping from Astana's PC to Almaty's PC

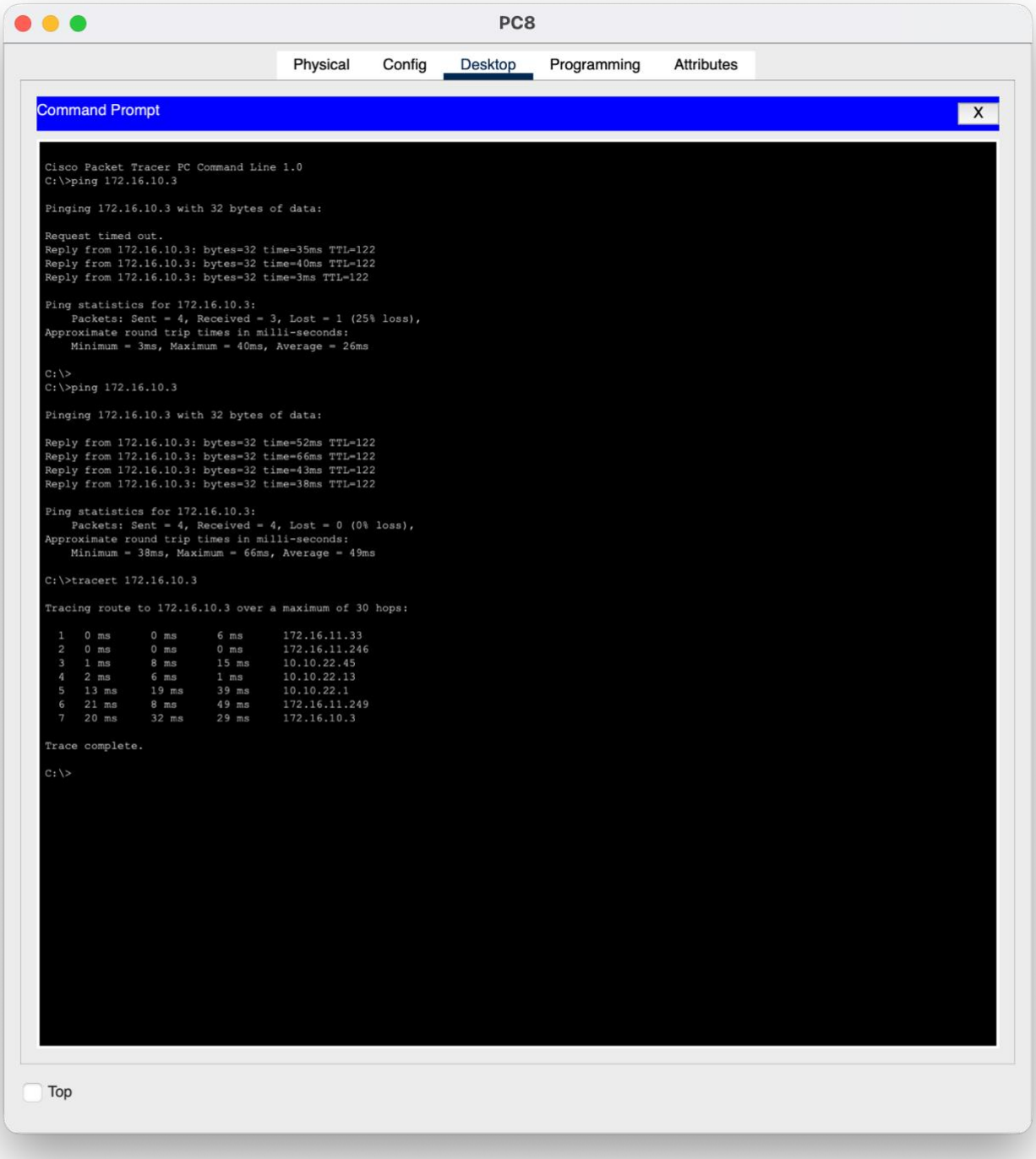


Figure D3 - Successful ping from Astana to Almaty and traceroute

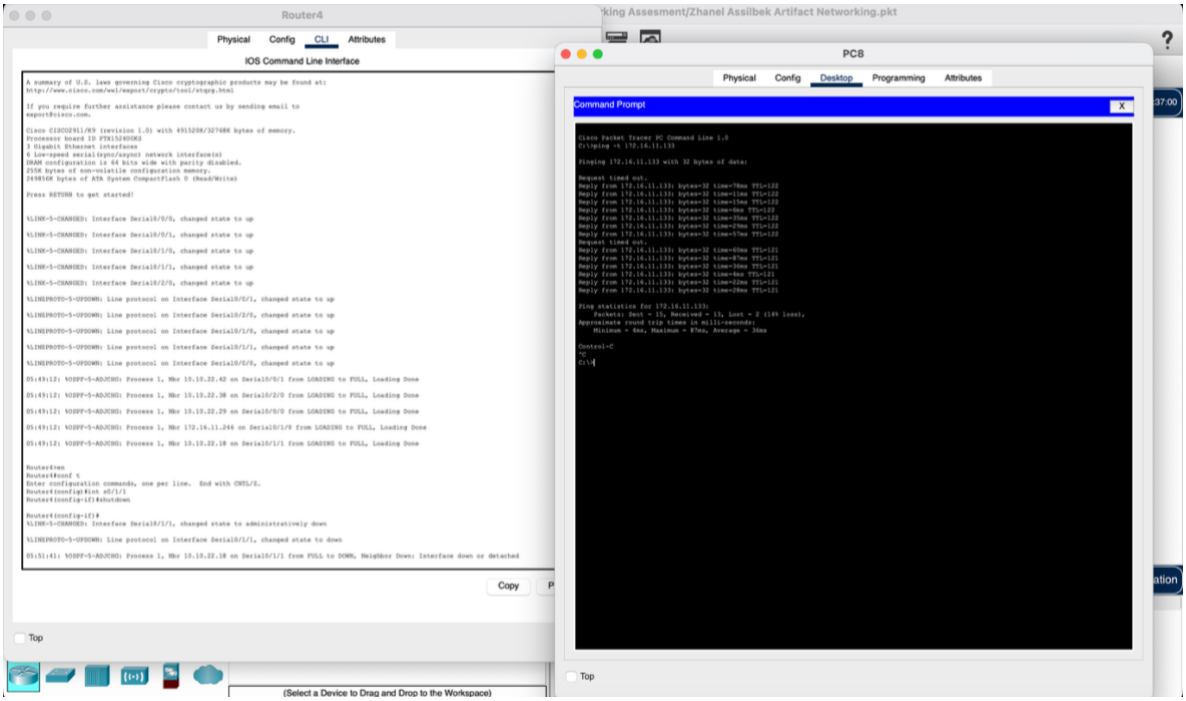


Figure D4 - Path Changing

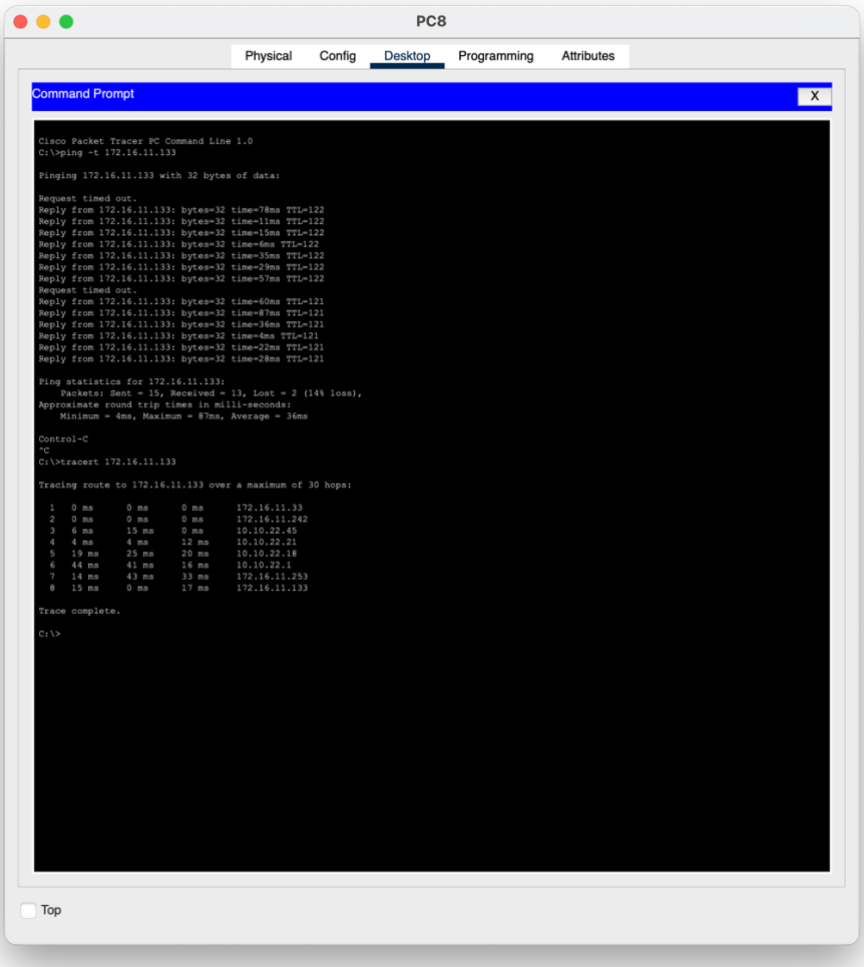


Figure D5 - Changed Traceroute

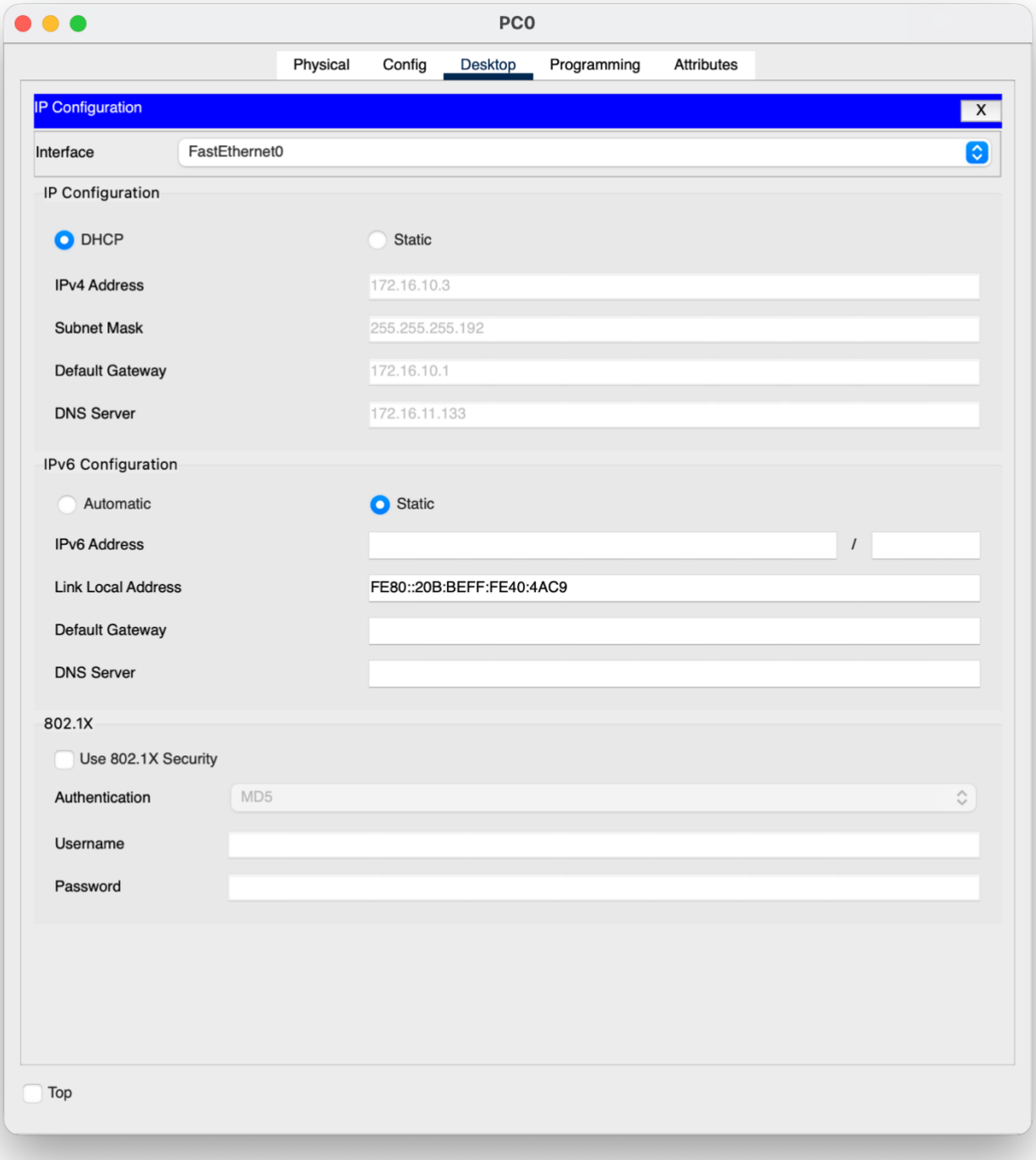


Figure D6 - Successful DHCP in Astana

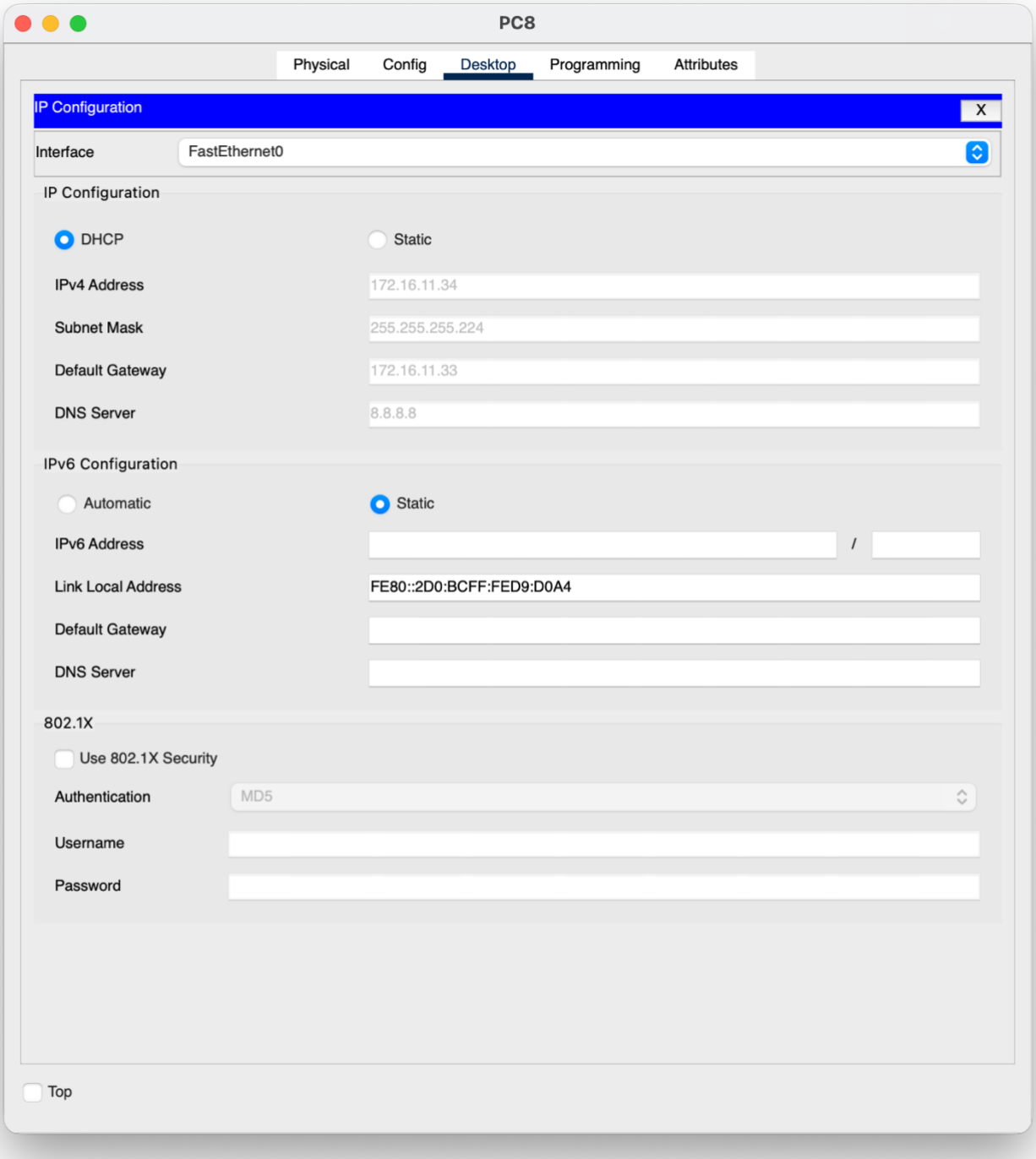


Figure D7 - Successful DHCP in Almaty

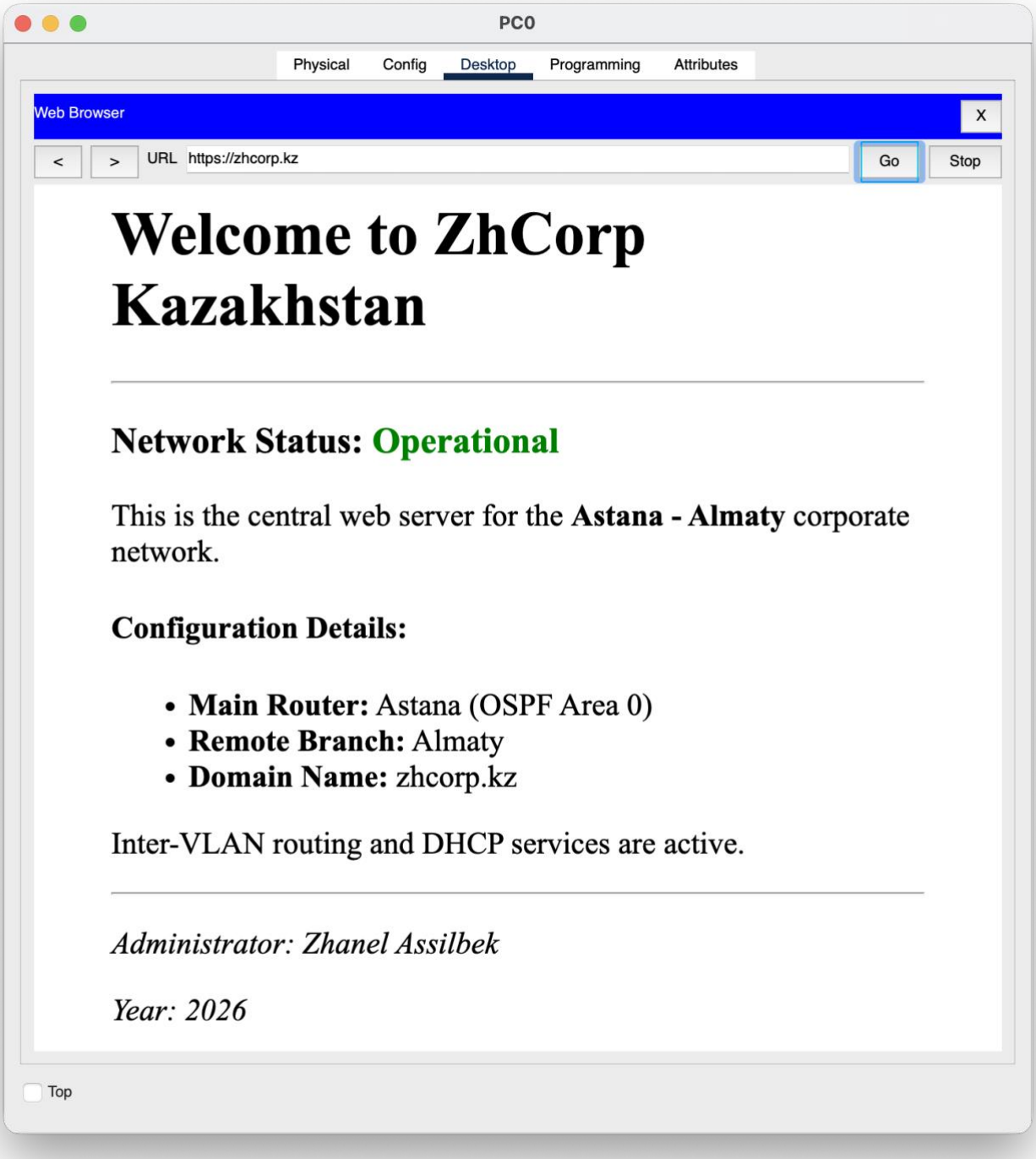


Figure D8 - Web-Server testing in Astana

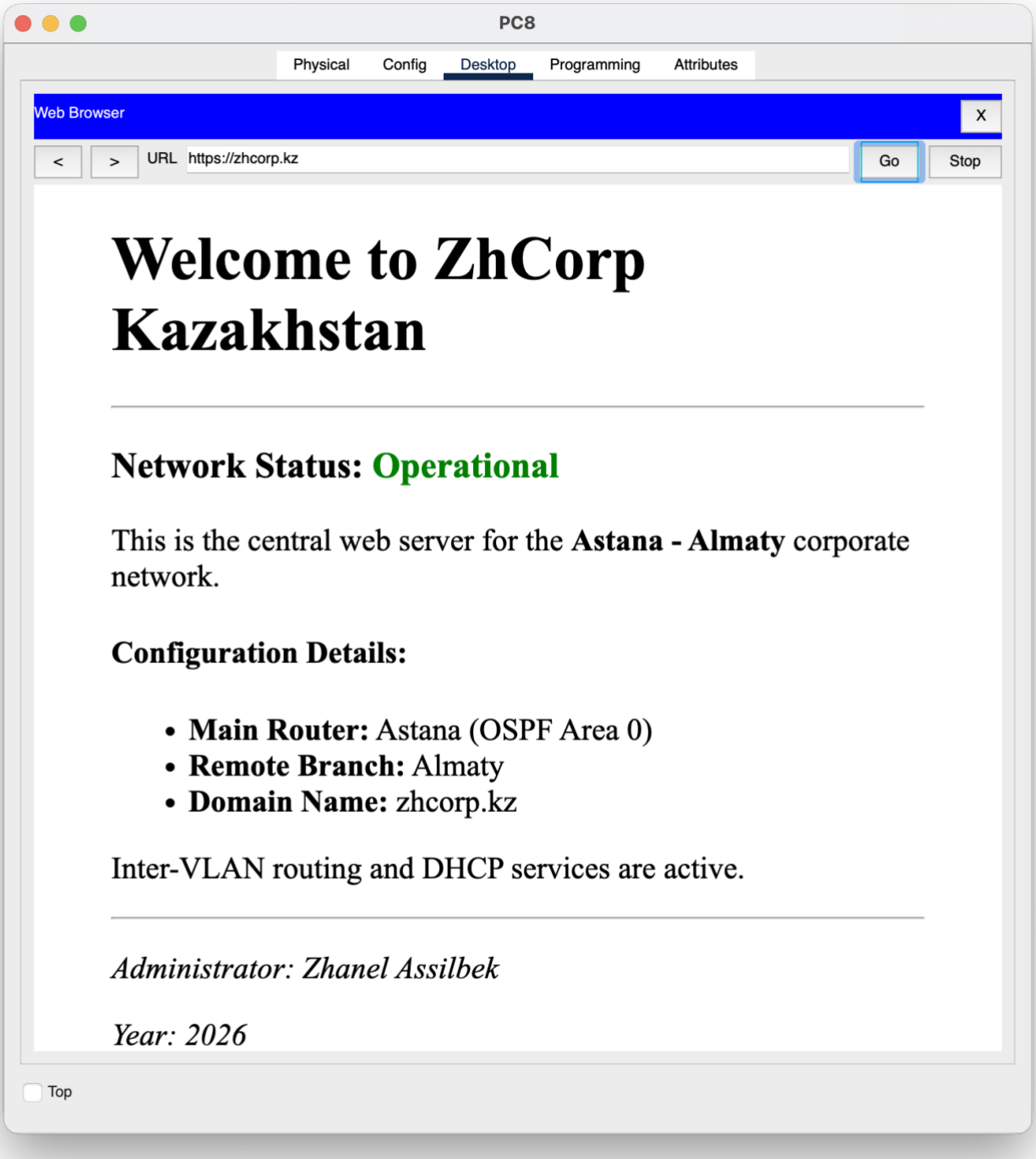


Figure D9 - Web-Server testing in Almaty.

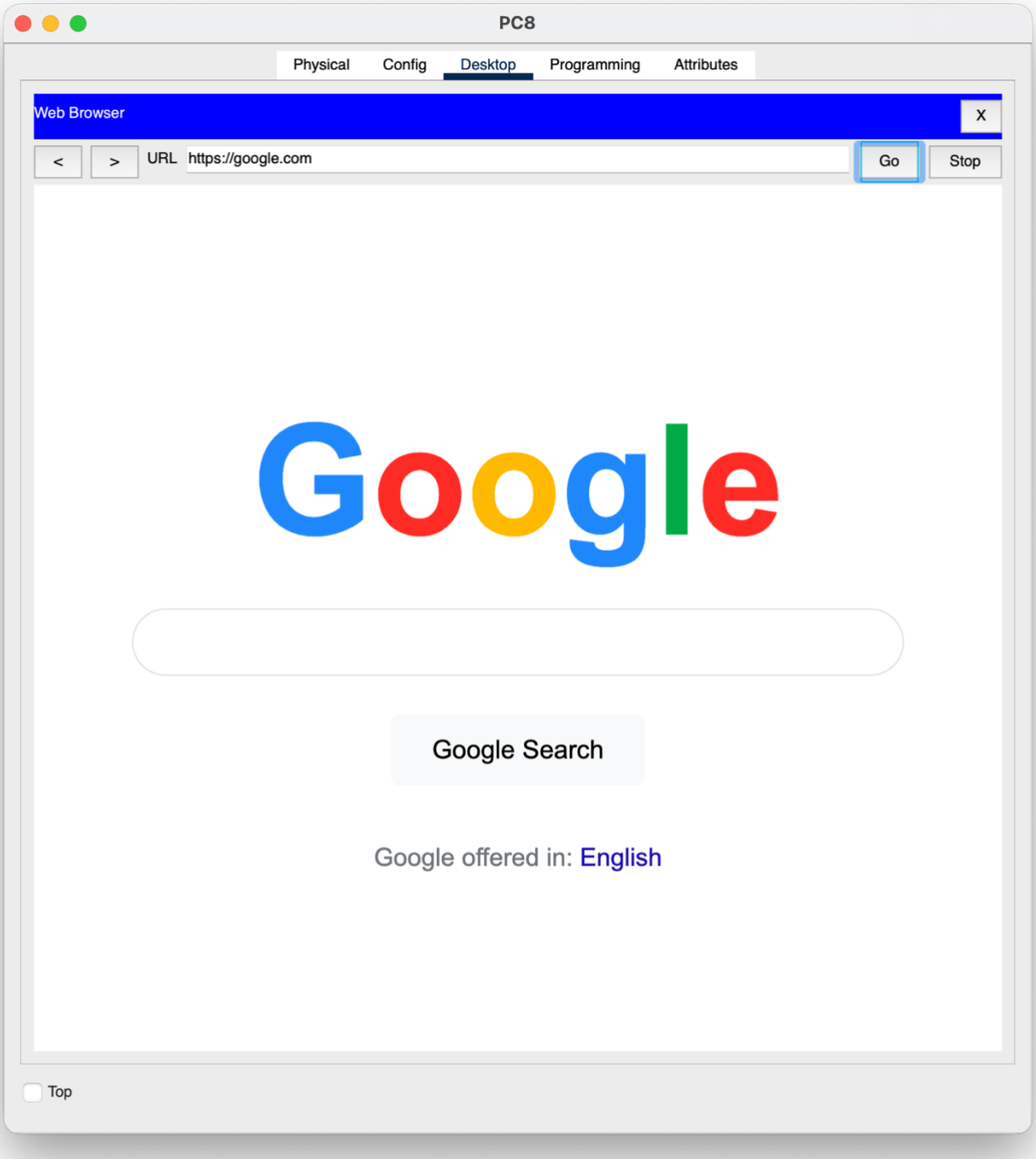


Figure D10 - Internet connection testing in Almaty

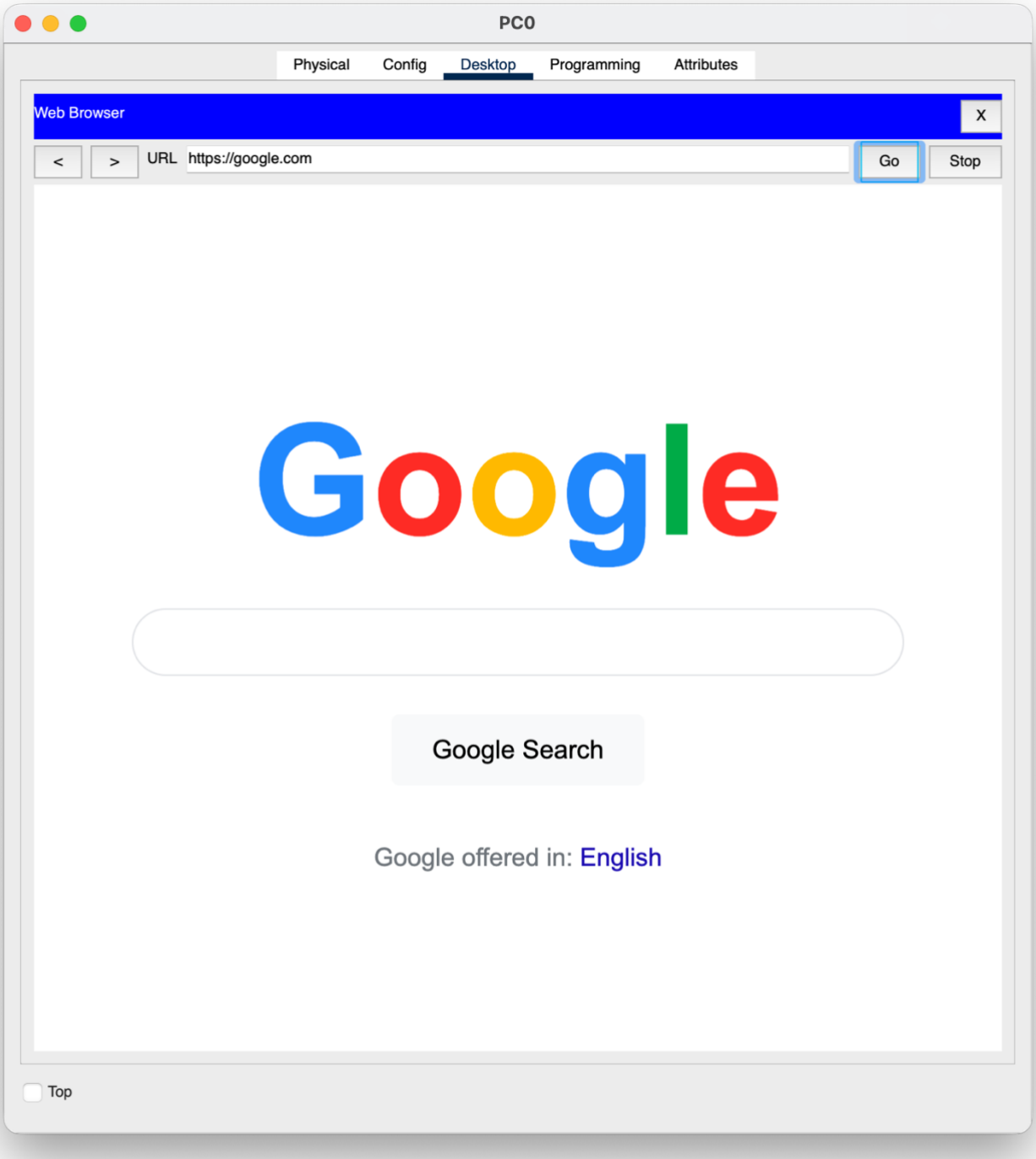


Figure D11 - Internet connection testing in Astana

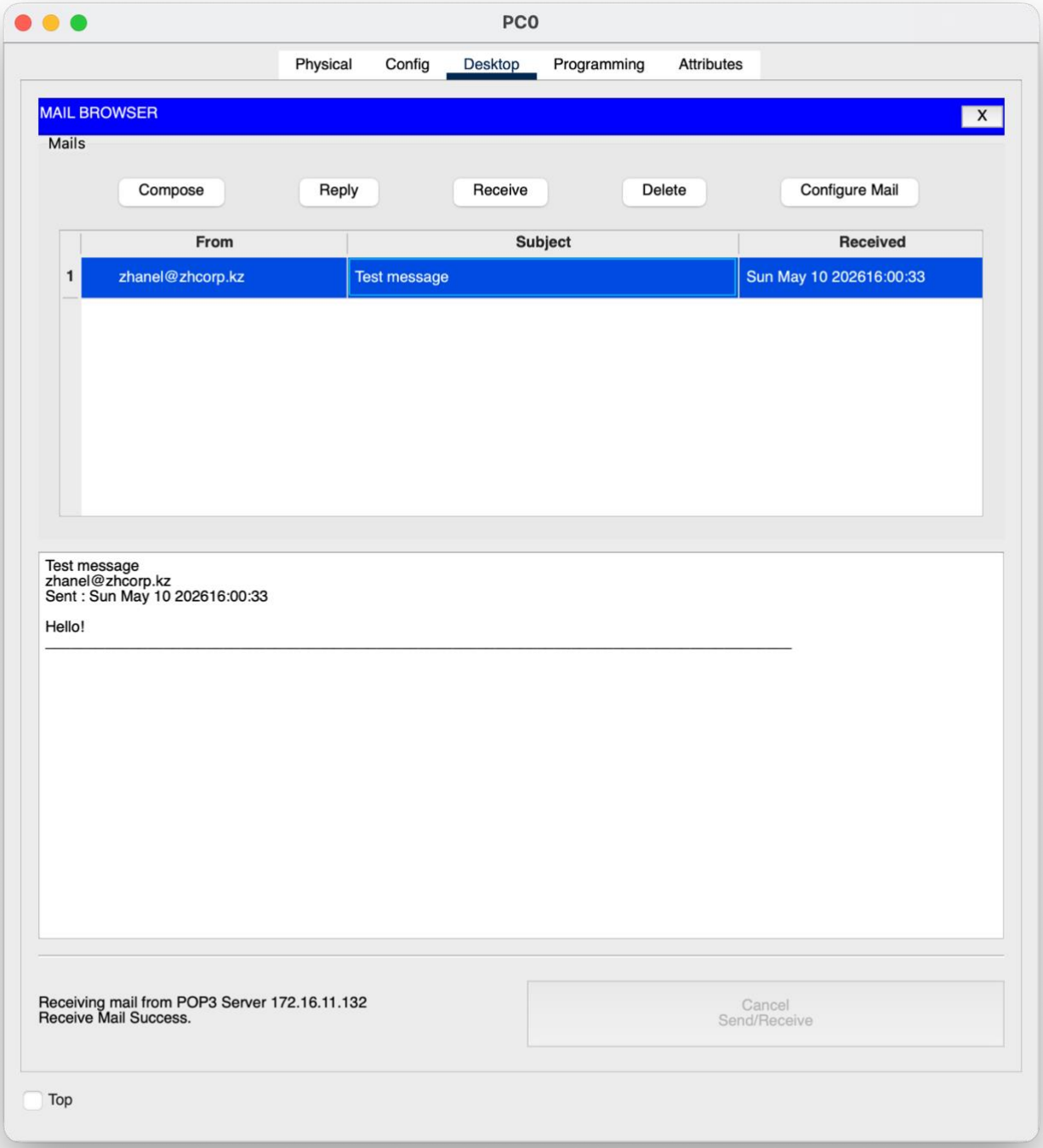


Figure D12 - Email server testing

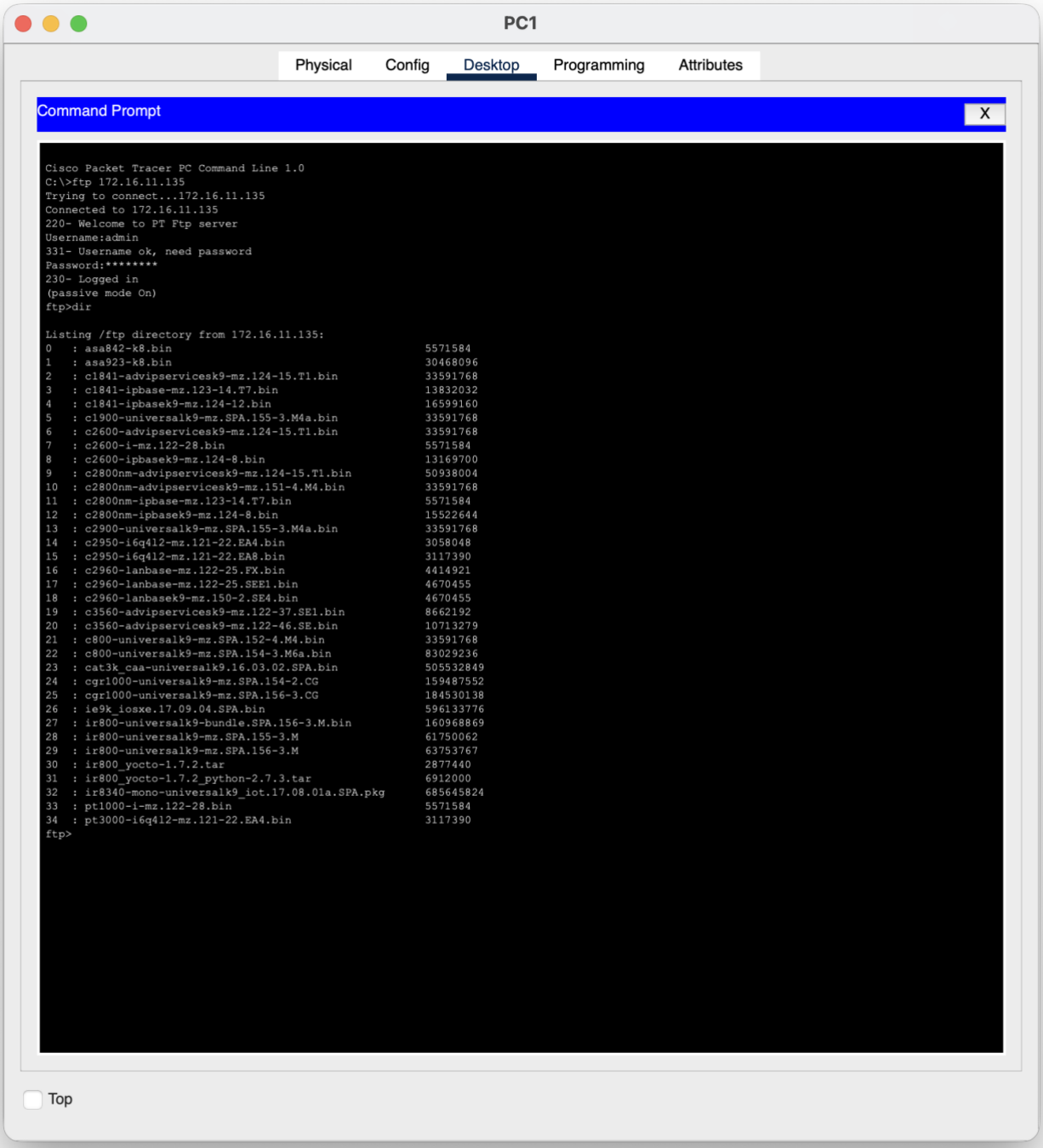


Figure D13 - FTP server testing in Astana

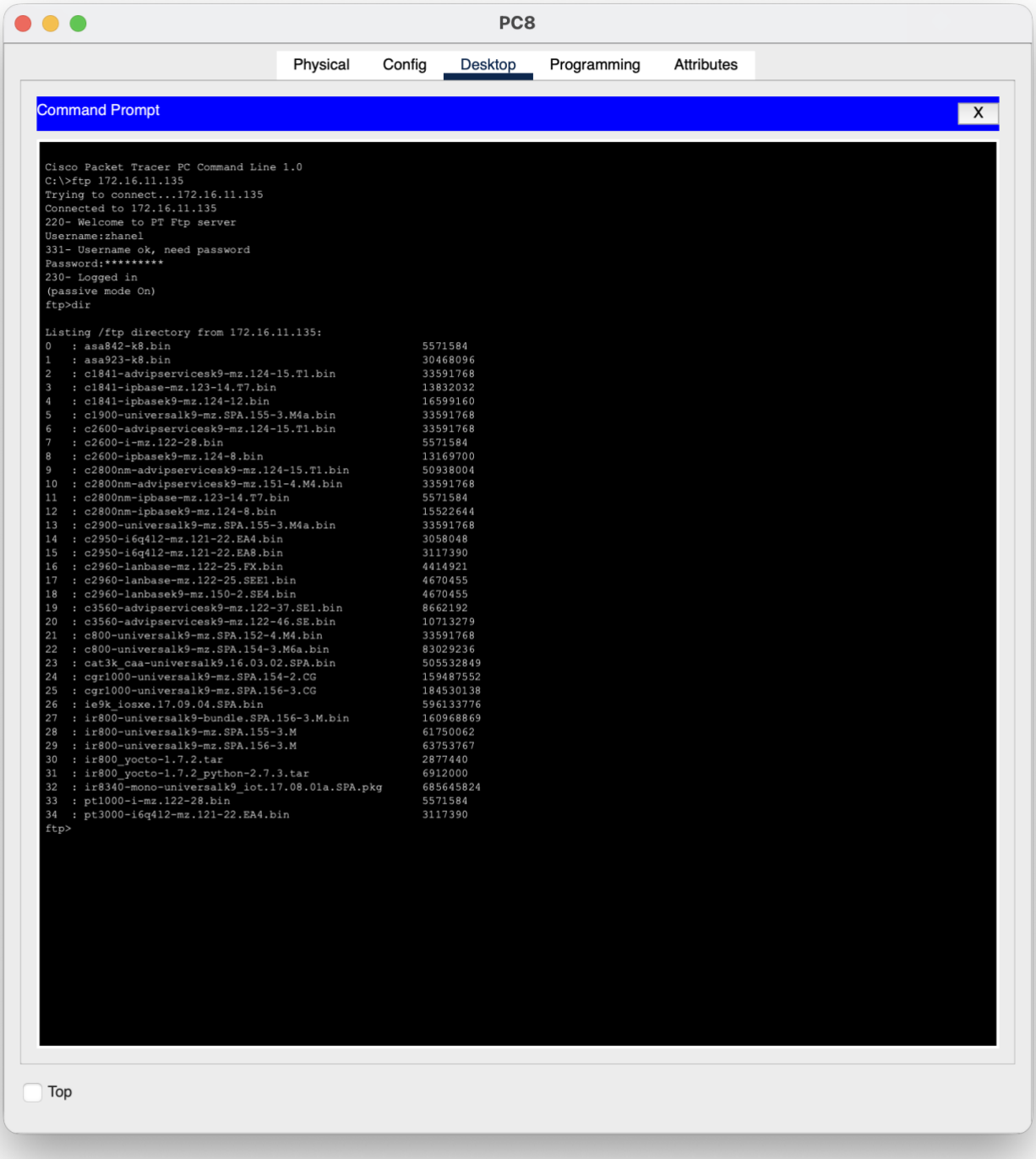


Figure D14 - FTP server testing in Almaty

```
[OK]
AST-Core-Active#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl110       10   110 P Active   local           172.16.10.61    172.16.10.1
Vl111       11   110 P Active   local           172.16.11.142   172.16.11.129
Vl120       20   110 P Active   local           172.16.10.125   172.16.10.65
Vl130       30   110 P Active   local           172.16.10.189   172.16.10.129
Vl140       40   110 P Active   local           172.16.10.253   172.16.10.193
Vl199       99   110 P Active   local           172.16.11.29    172.16.11.1
AST-Core-Active#
```

Figure D15 - HSRP testing on active core switch in Astana

```
AST-Core-Standby#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100   Standby  172.16.10.60    local           172.16.10.1
Vl11        11   100   Standby  172.16.11.141   local           172.16.11.129
Vl20        20   100   Standby  172.16.10.124   local           172.16.10.65
Vl30        30   100   Standby  172.16.10.188   local           172.16.10.129
Vl40        40   100   Standby  172.16.10.252   local           172.16.10.193
Vl99        99   100   Standby  172.16.11.2     local           172.16.11.1
AST-Core-Standby#
```

Figure D16 - HSRP testing on standby core switch in Astana

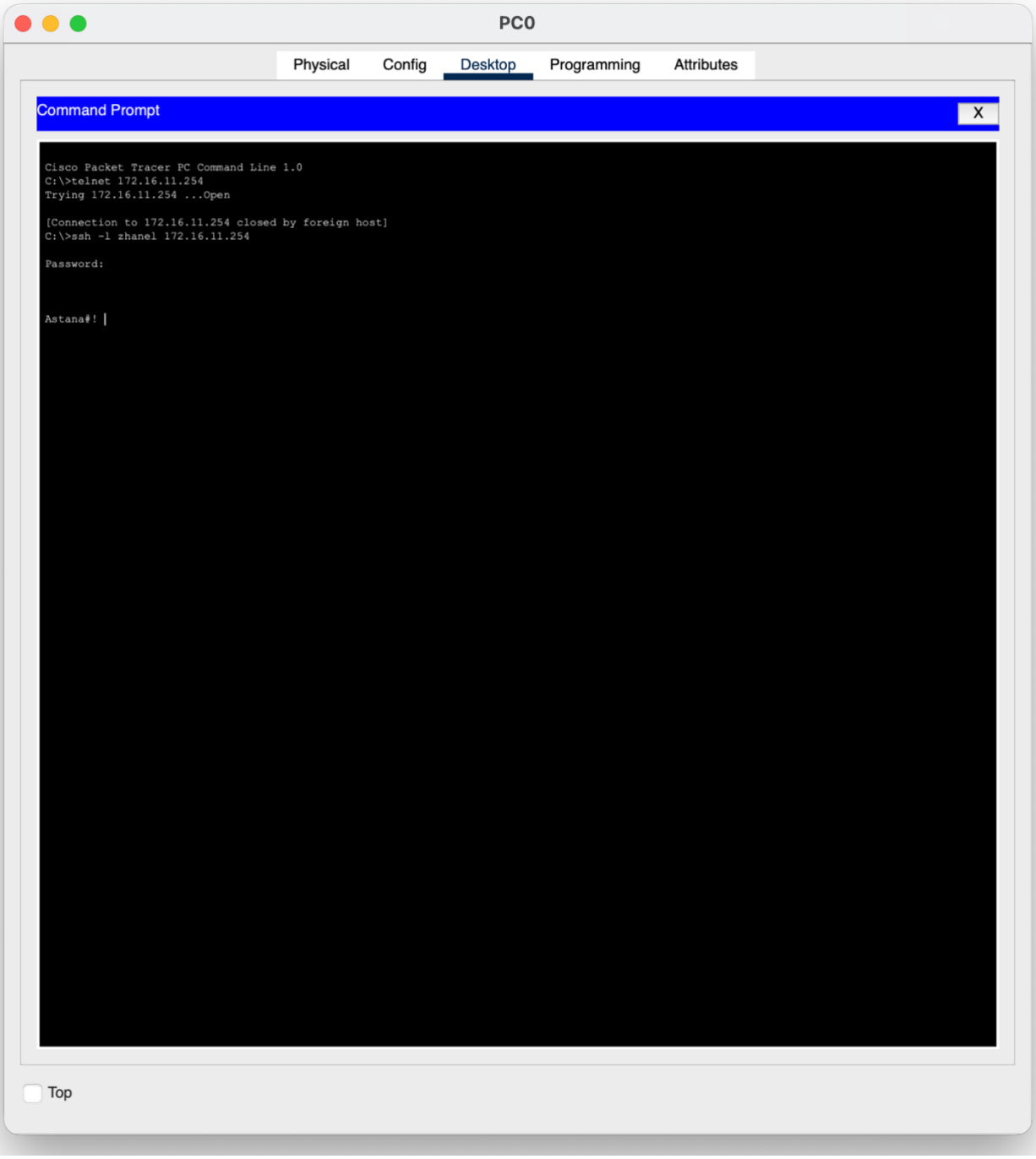


Figure D17 - SSH testing

```
Astana>en
Password:
Astana#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 85.22.10.2:1      172.16.10.10:1     8.8.8.8:1          8.8.8.8:1
icmp 85.22.10.2:2      172.16.10.10:2     8.8.8.8:2          8.8.8.8:2

Astana#
```

Figure D18 - NAT testing in Astana

```
Almaty>en
Password:
Almaty#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 85.22.10.6:1      172.16.11.55:1     8.8.8.8:1          8.8.8.8:1
icmp 85.22.10.6:2      172.16.11.55:2     8.8.8.8:2          8.8.8.8:2

Almaty#
```

Figure D19 - NAT testing in Almaty